

SS3 FURTHER MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: INEQUALITIES

1.1 INTRODUCTION

In mathematics, an **inequality** is a statement about the relative size or order of two objects. This week, we will extend our knowledge from linear inequalities to quadratic inequalities and the graphical representation of inequalities in two dimensions.

1.2 QUADRATIC INEQUALITIES

A **quadratic inequality** is one that can be written in the form:

$$ax^2 + bx + c > 0 \text{ (or } <, \leq, \geq \text{)}$$

where a, b, and c are real numbers and $a \neq 0$.

Solving Quadratic Inequalities

Method 1: Sign Chart / Test Regions (*Most reliable method*)

Steps:

1. **Standard Form:** Move all terms to one side (preferably the left) so that the inequality is compared to 0.
2. **Find Critical Values:** Solve the corresponding equation $ax^2 + bx + c = 0$ to find the roots (critical values). These are the points where the expression can change its sign.
3. **Create Regions:** Use the critical values to divide the number line into distinct regions (intervals).
4. **Test Points:** Pick one simple test point from each region and substitute it into the original inequality.
5. **Determine Solution:** If the test point makes the inequality true, then all numbers in that region are part of the solution. If it's false, that region is not part of the solution.

Example 1: Solve the inequality $x^2 - 5x + 6 > 0$

Step 1: Standard Form

Already in standard form: $x^2 - 5x + 6 > 0$

Step 2: Critical Values

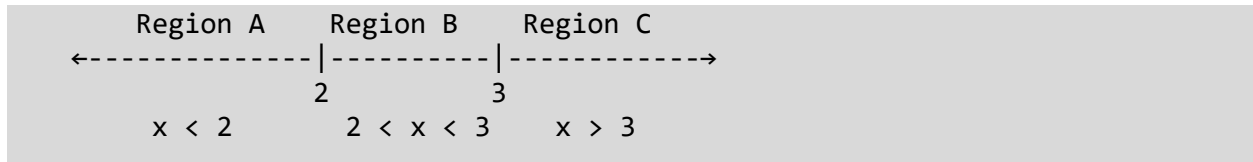
Solve $x^2 - 5x + 6 = 0$

Factoring: $(x - 2)(x - 3) = 0$

Critical values: $x = 2$ and $x = 3$

Step 3: Create Regions

These values divide the number line into three regions:

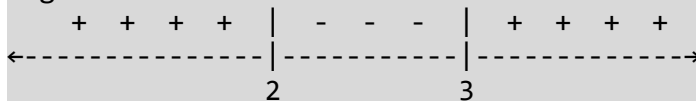


Step 4: Test Points

- **Region A** (Test $x = 0$):
 $(0)^2 - 5(0) + 6 > 0 \Rightarrow 6 > 0 \checkmark$ (**True**)
- **Region B** (Test $x = 2.5$):
 $(2.5)^2 - 5(2.5) + 6 > 0$
 $6.25 - 12.5 + 6 > 0$
 $-0.25 > 0 \times$ (**False**)
- **Region C** (Test $x = 4$):
 $(4)^2 - 5(4) + 6 > 0$
 $16 - 20 + 6 > 0$
 $2 > 0 \checkmark$ (**True**)

Step 5: Solution

Sign Chart:



The inequality is true for regions A and C.

Solution: $x < 2$ or $x > 3$

Method 2: Graphical Method (Quicker, more visual)

Steps:

1. **Find Critical Values:** Solve for the roots x_1 and x_2 .
2. **Sketch the Parabola:**
 - If $a > 0$ (the x^2 term is positive), the parabola is **U-shaped** (opens upwards)
 - If $a < 0$ (the x^2 term is negative), the parabola is **n-shaped** (opens downwards)
3. **Interpret the Graph:**

- The inequality $f(x) > 0$ asks: "Where is the graph **above** the x-axis?"
- The inequality $f(x) < 0$ asks: "Where is the graph **below** the x-axis?"

Example 2: Solve $3x - x^2 + 4 \geq 0$

Step 1: Standard Form

$$-x^2 + 3x + 4 \geq 0$$

Tip: To make factorization easier, multiply by -1 and **FLIP THE INEQUALITY SIGN**:

$$x^2 - 3x - 4 \leq 0$$

Step 2: Critical Values

$$\text{Solve } x^2 - 3x - 4 = 0$$

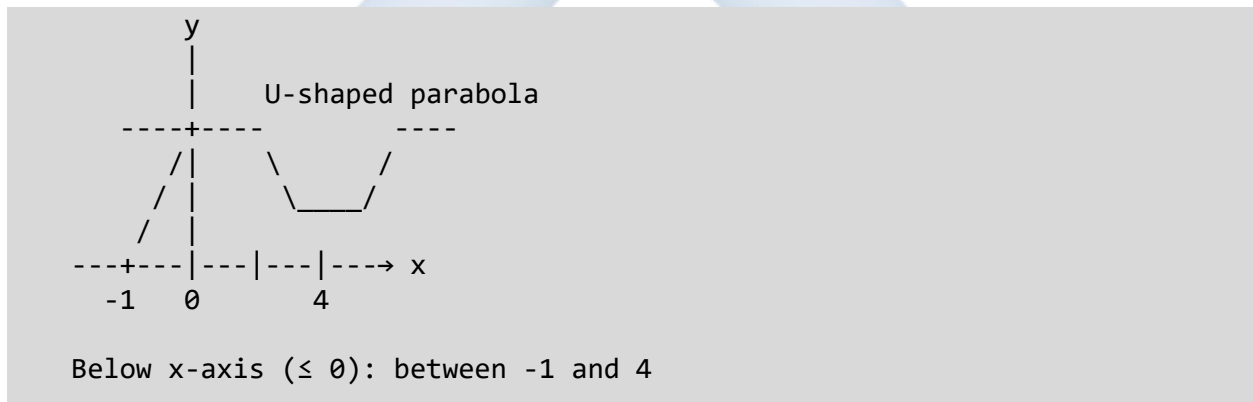
$$\text{Factoring: } (x - 4)(x + 1) = 0$$

Critical values: $x = 4$ and $x = -1$

Step 3: Sketch

We are solving $x^2 - 3x - 4 \leq 0$.

The a value is positive ($a = 1$), so it's a **U-shaped parabola**.



Step 4: Interpret

The graph is **below the x-axis** (i.e., ≤ 0) between the two roots.

Solution: $-1 \leq x \leq 4$

(Note: We use $\leq$ because the original inequality was $\geq$)

1.3 INEQUALITIES IN TWO DIMENSIONS

This involves graphing linear inequalities on a Cartesian (x-y) plane. The solution is not a set of numbers, but a **shaded region** (the "feasible region").

Method:

1. **Boundary Line:** Change the inequality (e.g., $y < 2x + 1$) to an equation ($y = 2x + 1$).
2. **Draw the Line:** Graph this line.

- Use a **solid line** (—) if the inequality is $\leq$ or $\geq$
 - Use a **dashed line** (- - -) if the inequality is $<$ or $>$
3. **Test a Point:** Choose a simple test point that is not on the line. The easiest is the **origin**, **(0, 0)**, if the line doesn't pass through it.
4. **Shade:**
- Substitute the test point into the original inequality
 - If the result is **True**, shade the entire region containing the test point
 - If the result is **False**, shade the region on the other side of the line

Example 3: Find the region that satisfies $2x + y < 4$

Step 1: Boundary Line

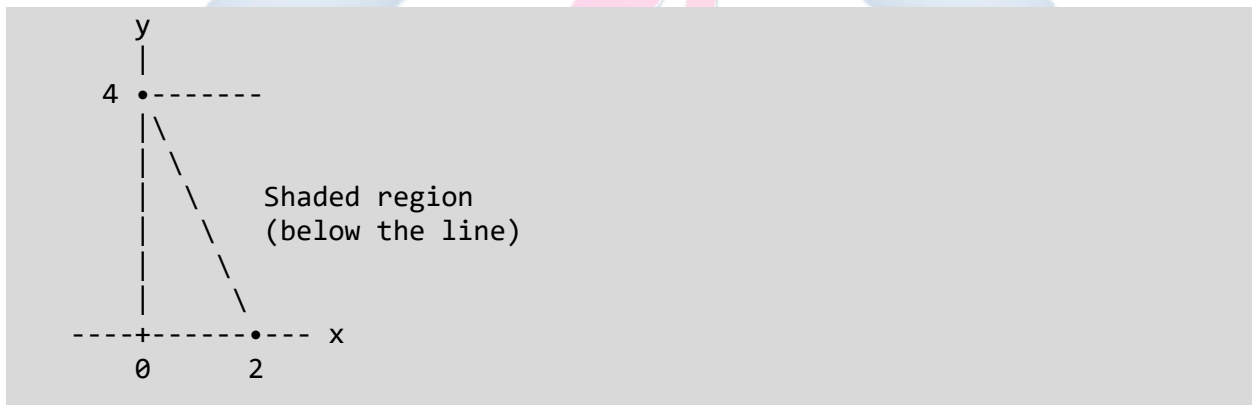
$$2x + y = 4$$

Find two points:

- When $x = 0$: $y = 4 \Rightarrow (0, 4)$
- When $y = 0$: $2x = 4 \Rightarrow x = 2 \Rightarrow (2, 0)$

Step 2: Draw the Line

Draw a **dashed line** through $(0, 4)$ and $(2, 0)$ (because the inequality is $<$, not $\leq$)



Step 3: Test Point

Use the origin $(0, 0)$:

$$2(0) + (0) < 4$$

$$0 < 4 \checkmark \text{ (True)}$$

Step 4: Shade

Shade the entire region containing the origin $(0, 0)$ — the region "**below**" the line.

Example 4: Show the region that satisfies $y \geq 1$, $x > -2$, and $y \leq -x + 3$

Line 1: $y \geq 1$

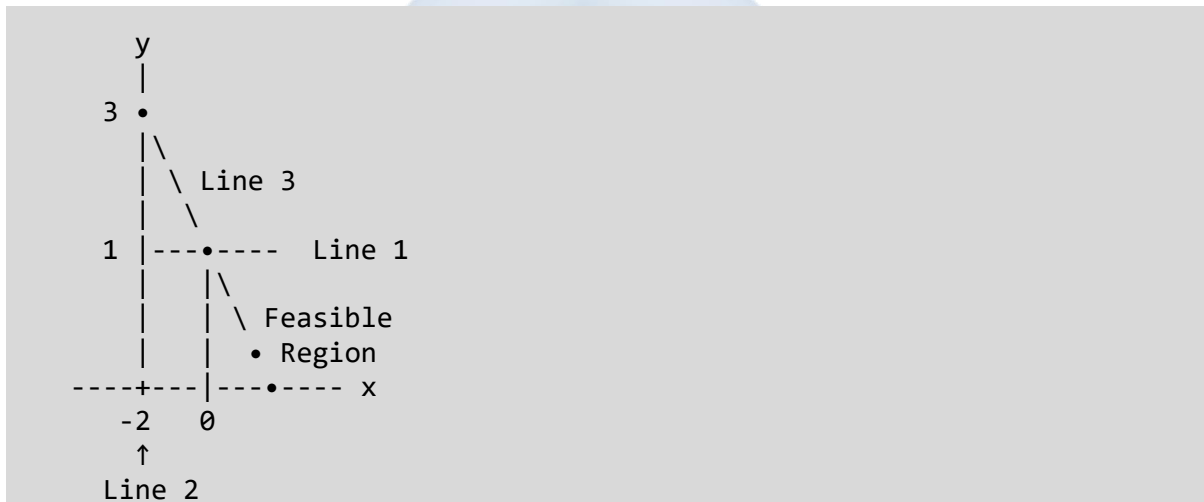
- A **solid horizontal line** at $y = 1$
- Test $(0, 0)$: $0 \geq 1$ ✗ (False)
- Shade **above** the line

Line 2: $x > -2$

- A **dashed vertical line** at $x = -2$
- Test $(0, 0)$: $0 > -2$ ✓ (True)
- Shade to the **right** of the line

Line 3: $y \leq -x + 3$

- A **solid line** ($y = -x + 3$)
- Passes through $(0, 3)$ and $(3, 0)$
- Test $(0, 0)$: $0 \leq -0 + 3 \Rightarrow 0 \leq 3$ ✓ (True)
- Shade **below** the line



Solution: The **feasible region** is the **triangle** formed by the intersection of all three shaded areas.

EVALUATION (WEEK 1)

1. Solve the inequality: $2x^2 + 5x - 3 < 0$
2. Find the range of values for k for which the equation $x^2 - kx + 4 = 0$ has real roots.
(Hint: $b^2 - 4ac \geq 0$)
3. Solve the inequality: $x^2 \geq 9$
4. Graphically represent the solution to $3x - 2y \leq 6$

ASSIGNMENT (WEEK 1)

1. Solve the inequality: $8 + 2x - x^2 \leq 0$
2. On a single graph, draw and shade the feasible region represented by the following inequalities:
 - $x + 2y \leq 10$
 - $3x + y \leq 15$
 - $x \geq 0$
 - $y \geq 0$
3. Find the coordinates of the vertices (corners) of the region in Q2.



WEEKS 2–3: MATRICES AND DETERMINANTS

2.1 MEANING OF MATRICES

A **matrix** is a rectangular array of numbers (or elements) arranged in rows and columns. Matrices are denoted by **capital letters**. The **order** (or dimension) of a matrix is given by $m \times n$, where:

- **m** = number of rows
- **n** = number of columns

Examples:

Matrix A (2×2 matrix - a "square matrix"):

$$A = \begin{bmatrix} 2 & 7 \\ -1 & 0 \end{bmatrix}$$

Matrix B (2×3 matrix):

$$B = \begin{bmatrix} 1 & 4 & 9 \\ 3 & 0 & -5 \end{bmatrix}$$

Special Matrices:

Zero Matrix (O): A matrix where all elements are zero.

$$O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Identity Matrix (I): A square matrix with 1s on the leading diagonal (top-left to bottom-right) and 0s elsewhere.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.2 ADDITION, SUBTRACTION, AND SCALAR MULTIPLICATION

⚠ Important Rule:

Matrices can only be **added or subtracted** if they have the **exact same order** (same number of rows and columns).

Operations:

Addition/Subtraction: Add or subtract the corresponding elements.

Scalar Multiplication: To multiply a matrix by a number (a scalar), multiply **every element** in the matrix by that number.

Example 1:

Let $A = \begin{bmatrix} 4 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} \begin{bmatrix} 5 & 6 \end{bmatrix}$

Find: (a) $A + B$ (b) $A - B$ (c) $3A$

(a) $A + B$

$$A + B = \begin{bmatrix} 4+0 & 8+2 \\ 3+5 & -1+6 \end{bmatrix} = \begin{bmatrix} 4 & 10 \\ 8 & 5 \end{bmatrix}$$

(b) $A - B$

$$A - B = \begin{bmatrix} 4-0 & 8-2 \\ 3-5 & -1-6 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ -2 & -7 \end{bmatrix}$$

(c) $3A$

$$3A = 3 \times \begin{bmatrix} 4 & 8 \\ 3 & -1 \end{bmatrix} = \begin{bmatrix} 3(4) & 3(8) \\ 3(3) & 3(-1) \end{bmatrix} = \begin{bmatrix} 12 & 24 \\ 9 & -3 \end{bmatrix}$$

2.3 MATRIX MULTIPLICATION

Rule:

To multiply two matrices, $A \times B$, the **number of columns in A** must equal the **number of rows in B**.

If A is $(m \times n)$ and B is $(n \times p)$, the resulting matrix AB will be $(m \times p)$.

Method: "Row-by-Column"

To find the element in **row i**, **column j** of the answer:

1. Take the elements of **row i** from matrix A
2. Multiply them by the elements of **column j** from matrix B
3. **Sum them up**

⚠ Important Note:

Matrix multiplication is NOT commutative.

$AB \neq BA$ (in general)

Example 2:

Let $C = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 6 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} 7 & 8 \end{bmatrix}$

Find **CD**.

Order check: $(2 \times 2) \times (2 \times 2) \rightarrow (2 \times 2) \checkmark$ **This is possible.**

Calculation:

$$CD = \begin{bmatrix} (1 \cdot 5 + 2 \cdot 7) & (1 \cdot 6 + 2 \cdot 8) \\ (3 \cdot 5 + 4 \cdot 7) & (3 \cdot 6 + 4 \cdot 8) \end{bmatrix}$$

$$= \begin{bmatrix} (5 + 14) & (6 + 16) \\ (15 + 28) & (18 + 32) \end{bmatrix}$$

$$= \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Visual representation:

$$\text{Row 1} \times \text{Col 1: } \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 7 \end{bmatrix} = 1(5) + 2(7) = 19$$

$$\text{Row 1} \times \text{Col 2: } \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 8 \end{bmatrix} = 1(6) + 2(8) = 22$$

$$\text{Row 2} \times \text{Col 1: } \begin{bmatrix} 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 7 \end{bmatrix} = 3(5) + 4(7) = 43$$

$$\text{Row 2} \times \text{Col 2: } \begin{bmatrix} 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 8 \end{bmatrix} = 3(6) + 4(8) = 50$$

2.4 MATRICES AS LINEAR TRANSFORMATIONS

A matrix can be used to **"transform"** a point (x, y) or a shape in a 2D plane.

We write the point as a **column vector** $\begin{bmatrix} x \\ y \end{bmatrix}$

The new point (x', y') is found by:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \text{Transformation Matrix} \times \begin{bmatrix} x \\ y \end{bmatrix}$$

Standard Transformation Matrices (MEMORIZE THESE!)

Transformation	Matrix	Notation
Reflection in x-axis ($y = 0$)	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	R_x
Reflection in y-axis ($x = 0$)	$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$	R_y
Reflection in line $y = x$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$R_{y=x}$
Reflection in line $y = -x$	$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$	$R_{y=-x}$
Rotation 90° anti-clockwise (origin)	$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$	Rot_{90}
Rotation 180° (origin)	$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$	Rot_{180}
Enlargement, scale factor k (origin)	$\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$	E

Example 3:

Find the image of the point $P(3, -2)$ under a 90° anti-clockwise rotation about the origin.

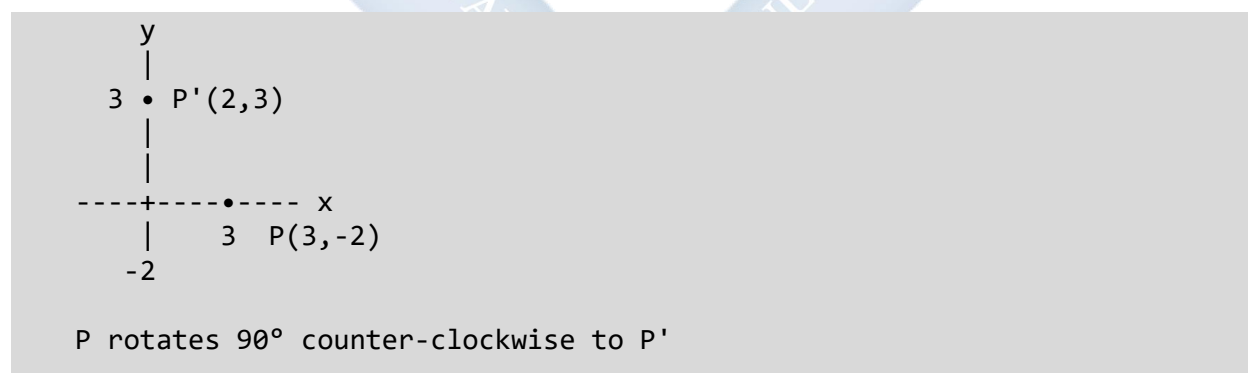
Solution:

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} 3 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} (0 \cdot 3) + (-1 \cdot -2) \\ (1 \cdot 3) + (0 \cdot -2) \end{bmatrix} = \begin{bmatrix} 0 + 2 \\ 3 + 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

The new point is $P'(2, 3)$

Visual representation:



2.5 DETERMINANTS (2×2)

The **determinant** is a single number associated with a square matrix.

For a 2×2 matrix: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

The determinant is:

$$|A| = \det(A) = ad - bc$$

Visual memory aid:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow (a \times d) - (b \times c)$$

↘ ↗

Singular Matrix:

A matrix is "**singular**" if its determinant is **0**. A singular matrix has **no inverse**.

Inverse of a 2×2 Matrix (A^{-1}):

Steps:

1. Find the determinant, $|A|$
2. **Swap** the elements on the leading diagonal ($a \leftrightarrow d$)
3. **Change the sign** of the elements on the other diagonal ($b \rightarrow -b, c \rightarrow -c$)
4. Multiply the new matrix by $1/|A|$

Formula:

$$A^{-1} = 1/(ad-bc) \times \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 4:

Find the inverse of $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Step 1: Find $|C|$

$$|C| = (1)(4) - (2)(3) = 4 - 6 = -2$$

Step 2: Apply the inverse formula

$$C^{-1} = 1/(-2) \times \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$$

Verification: $C \times C^{-1} = I \checkmark$

2.6 DETERMINANTS (3×3)

We use the **Method of Cofactors** (or Sarrus' Rule).

Method of Cofactors:

1. Sign Matrix Pattern:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

2. Choose any single row or column (usually the **first row**)

3. For each element in that row:

- Multiply the element by its "**sign**" (+ or -)
- Multiply by the **determinant of the 2×2 matrix** left when you "cover up" that element's row and column

Example 5:

Find the determinant of $M = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$

Expand along Row 1:

$$|M| = +1 \cdot \begin{vmatrix} 4 & 5 \\ 0 & 6 \end{vmatrix} - 2 \cdot \begin{vmatrix} 0 & 5 \\ 1 & 6 \end{vmatrix} + 3 \cdot \begin{vmatrix} 0 & 4 \\ 1 & 0 \end{vmatrix}$$

Calculate each 2×2 determinant:

$$|M| = 1 \cdot [(4 \cdot 6) - (5 \cdot 0)] - 2 \cdot [(0 \cdot 6) - (5 \cdot 1)] + 3 \cdot [(0 \cdot 0) - (4 \cdot 1)]$$

$$= 1 \cdot (24 - 0) - 2 \cdot (0 - 5) + 3 \cdot (0 - 4)$$

$$= 1(24) - 2(-5) + 3(-4)$$

$$= 24 + 10 - 12$$

$$= 22$$

$$|M| = 22$$

2.7 SOLUTION OF SIMULTANEOUS EQUATIONS (CRAMER'S RULE)

Cramer's Rule uses **determinants** to solve systems of linear equations.

For a 2×2 System:

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$

Steps:

1. **Find Δ** (determinant of coefficient matrix):

$$\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

2. **Find Δ_x** (replace x-column with constants):

$$\Delta_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}$$

3. **Find Δ_y** (replace y-column with constants):

$$\Delta_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

4. **Solution:**

$$x = \Delta_x / \Delta \quad \text{and} \quad y = \Delta_y / \Delta$$

Example 6:

Solve: $2x + y = 7$

$$3x - 2y = 0$$

Step 1: Find Δ

$$\Delta = \begin{vmatrix} 2 & 1 \\ 3 & -2 \end{vmatrix} = (2)(-2) - (1)(3) = -4 - 3 = -7$$

Step 2: Find Δ_x (replace x-column)

$$\Delta_x = \begin{vmatrix} 7 & 1 \\ 0 & -2 \end{vmatrix} = (7)(-2) - (1)(0) = -14$$

Step 3: Find Δ_y (replace y-column)

$$\Delta_y = \begin{vmatrix} 2 & 7 \\ 3 & 0 \end{vmatrix} = (2)(0) - (7)(3) = -21$$

Step 4: Calculate x and y

$$x = \Delta_x / \Delta = (-14) / (-7) = 2$$

$$y = \Delta_y / \Delta = (-21) / (-7) = 3$$

✓ **Solution:** $x = 2, y = 3$

For a 3×3 System:

The principle is **identical**:

$$x = \Delta_x / \Delta, \quad y = \Delta_y / \Delta, \quad z = \Delta_z / \Delta$$

where:

- Δ is the 3×3 determinant of the coefficients
- $\Delta_x, \Delta_y, \Delta_z$ are the determinants found by replacing the x, y, or z columns (respectively) with the constants

EVALUATION (WEEKS 2–3)

1. If $A = \begin{bmatrix} -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 1 \end{bmatrix}$, find $2A - B$. $\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -4 \end{bmatrix}$
2. If $P = \begin{bmatrix} 2 & 1 \end{bmatrix}$, find P^{-1} . $\begin{bmatrix} 5 & 3 \end{bmatrix}$
3. Find the determinant of $M = \begin{bmatrix} 1 & 3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 4 \end{bmatrix} \begin{bmatrix} -1 & 5 & 0 \end{bmatrix}$
4. What transformation is represented by the matrix $\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix}$?
5. Use Cramer's rule to solve: $5x + 2y = 4$ and $7x + 3y = 5$.

ASSIGNMENT (WEEKS 2–3)

1. Let $P = \begin{bmatrix} 1 & -2 \end{bmatrix}$ and $Q = \begin{bmatrix} 4 & 1 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 & 3 \end{bmatrix}$
 - (a) Can you find PQ ? If so, find it.
 - (b) Can you find QP ? If so, find it.
2. Find the image of the triangle with vertices $A(1,1)$, $B(3,1)$, $C(3,5)$ under the transformation given by the matrix:

$$T = \begin{bmatrix} 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Draw the original and new triangles on a graph.

3. Use Cramer's rule to solve the 3×3 system:

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$x + 2y - 3z = -4$$

WEEK 4: PARTIAL FRACTIONS

4.1 BASIC DEFINITION

A **rational function** is a fraction where the numerator and denominator are both polynomials:

$$P(x) / Q(x)$$

Partial fractions is a technique used to decompose a complex rational function into a sum of simpler fractions. This is a vital skill for more advanced calculus (integration).

Types of Rational Functions:

Proper Rational Function:

The degree of the numerator is **less than** the degree of the denominator.

Example: $(x + 1)/(x^2 + 3x + 2) \rightarrow \text{degree } 1 < \text{degree } 2 \checkmark$

We can use partial fractions directly.

Improper Rational Function:

The degree of the numerator is **equal to or greater than** the degree of the denominator.

Example: $(x^3 + 1)/(x^2 + 3x + 2) \rightarrow \text{degree } 3 > \text{degree } 2$

You must perform polynomial long division first to get:

Polynomial + Proper Rational Function

4.2 CASE 1: DENOMINATOR WITH DISTINCT LINEAR FACTORS

Form:

$$\frac{P(x)}{(x-a)(x-b)(x-c)} = \frac{A}{(x-a)} + \frac{B}{(x-b)} + \frac{C}{(x-c)}$$

We must find the constants **A, B, and C**.

Example 1: Decompose $(7x - 1)/[(x - 3)(x + 1)]$ into partial fractions.

Method A: Equating Coefficients

Step 1: Set up the identity

$$\frac{7x - 1}{(x-3)(x+1)} \equiv \frac{A}{(x-3)} + \frac{B}{(x+1)}$$

Step 2: Multiply by the common denominator

$$7x - 1 \equiv A(x + 1) + B(x - 3)$$

Step 3: Expand and collect like terms

$$7x - 1 \equiv Ax + A + Bx - 3B$$

$$7x - 1 \equiv (A + B)x + (A - 3B)$$

Step 4: Equate coefficients

$$\text{Coefficient of } x: A + B = 7 \quad \dots (1)$$

$$\text{Constant term: } A - 3B = -1 \quad \dots (2)$$

Step 5: Solve the simultaneous equations

$$\text{From (1): } A = 7 - B$$

Substitute into (2):

$$(7 - B) - 3B = -1$$

$$7 - B - 3B = -1$$

$$7 - 4B = -1$$

$$-4B = -8$$

$$B = 2$$

Substitute $B = 2$ into (1):

$$A + 2 = 7$$

$$A = 5$$

✓ Solution:

$$\frac{7x - 1}{(x-3)(x+1)} = \frac{5}{(x-3)} + \frac{2}{(x+1)}$$

Method B: "Cover-Up" Method (Heaviside's Method)

✂ Much faster for distinct linear factors!

Start from: $7x - 1 \equiv A(x + 1) + B(x - 3)$

To find A:

1. "Cover up" its denominator ($x - 3$)
2. Let $x = 3$ in the remaining expression:

Let $x = 3$:

$$7(3) - 1 = A(3 + 1) + B(0)$$

$$21 - 1 = 4A$$

$$20 = 4A$$

$$A = 5 \checkmark$$

To find B:

1. "Cover up" its denominator ($x + 1$)
2. Let $x = -1$ in the remaining expression:

Let $x = -1$:

$$7(-1) - 1 = A(0) + B(-1 - 3)$$

$$-7 - 1 = -4B$$

$$-8 = -4B$$

$$B = 2 \checkmark$$

✓ Solution:

$$\frac{7x - 1}{(x-3)(x+1)} = \frac{5}{(x-3)} + \frac{2}{(x+1)}$$

Visual representation:

$$\frac{7x - 1}{(x-3)(x+1)}$$

↓ ↓
→ Cover up $(x+1)$, let $x = -1 \rightarrow B = 2$
→ Cover up $(x-3)$, let $x = 3 \rightarrow A = 5$

4.3 CASE 2: DENOMINATOR WITH REPEATED LINEAR FACTORS**Form:**

A factor like $(x - a)^n$ requires **n partial fractions**, one for each power.

$$\frac{P(x)}{(x-a)^2(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-a)^2} + \frac{C}{(x-b)}$$

⚠ Important: The cover-up method will **only work** for:

- The **highest power** of the repeated factor (B)
- The **distinct factor** (C)

You must use **substitution** or **equating coefficients** to find A.

Example 2: Decompose $(x^2 + 5x + 7)/[(x - 1)(x + 2)^2]$ into partial fractions.

Step 1: Set up the form

$$\frac{x^2 + 5x + 7}{(x-1)(x+2)^2} = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2}$$

Step 2: Multiply by LCD

$$x^2 + 5x + 7 \equiv A(x + 2)^2 + B(x - 1)(x + 2) + C(x - 1)$$

Step 3: Use Cover-Up (where possible)

Find A (let $x = 1$):

$$(1)^2 + 5(1) + 7 = A(1 + 2)^2 + B(0) + C(0)$$

$$1 + 5 + 7 = A(3)^2$$

$$13 = 9A$$

$$A = 13/9 \checkmark$$

Find C (let $x = -2$):

$$(-2)^2 + 5(-2) + 7 = A(0) + B(0) + C(-2 - 1)$$

$$4 - 10 + 7 = C(-3)$$

$$1 = -3C$$

$$C = -1/3 \checkmark$$

Step 4: Find B (No cover-up available)

Use substitution with a simple value, $x = 0$:

$$(0)^2 + 5(0) + 7 = A(0 + 2)^2 + B(0 - 1)(0 + 2) + C(0 - 1)$$

$$7 = 4A - 2B - C$$

Substitute known values ($A = 13/9$, $C = -1/3$):

$$7 = 4(13/9) - 2B - (-1/3)$$

$$7 = 52/9 - 2B + 1/3$$

Multiply by 9:

$$63 = 52 - 18B + 3$$

$$63 = 55 - 18B$$

$$8 = -18B$$

$$B = -4/9 \checkmark$$

✓ **Solution:**

$$\frac{x^2 + 5x + 7}{(x-1)(x+2)^2} = \frac{13/9}{(x-1)} + \frac{-4/9}{(x+2)} + \frac{-1/3}{(x+2)^2}$$

EVALUATION (WEEK 4)

1. What is the difference between a "proper" and "improper" rational function? What must be done before decomposing an improper function?
2. Set up the correct partial fraction form for $5x/[(x+3)^3(x-1)]$.
(Do not solve for the constants)
3. Decompose $(10x+4)/(x^2-4)$ into partial fractions.
(Hint: Factorize the denominator first)

ASSIGNMENT (WEEK 4)

1. Decompose $(x-11)/[(x-1)(x+3)]$ using **both** the "equating coefficients" and "cover-up" methods.
2. Find the partial fractions for $(2x^2+x+1)/[(x-1)^2(x+1)]$.

WEEKS 5–6: PROBABILITY DISTRIBUTIONS

5.1 INTRODUCTION

A **Random Variable** (usually X) is a variable whose value is the numerical outcome of a random phenomenon.

Types of Random Variables:

Discrete Random Variable:

Can only take **specific, countable** values.

Example: Number of heads in 3 coin flips $\rightarrow \{0, 1, 2, 3\}$

Continuous Random Variable:

Can take **any value** within a given range.

Example: Height of a student $\rightarrow 170\text{cm}, 170.1\text{cm}, 170.11\text{cm}, \dots$

A **Probability Distribution** is a function that describes the probability of every possible outcome for a random variable.

5.2 BINOMIAL DISTRIBUTION [$X \sim B(n, p)$]

Context:

Used for **discrete data**, when we count the number of "**successes**" in a **fixed number** of independent trials.

Conditions (B.I.N.S.):

- ✓ **Binary**: There are only **two possible outcomes** for each trial (e.g., Success/Failure, Head/Tail, Defective/Not-Defective)
- ✓ **Independent**: The outcome of one trial **does not affect** the next
- ✓ **Number**: The **number of trials, n** , is fixed in advance
- ✓ **Success**: The **probability of success, p** , is the same for every trial (The probability of failure is **$q = 1 - p$**)

Formula:

The probability of **exactly x successes** in **n trials** is:

$$P(X = x) = \frac{(n)!}{(x)!(n-x)!} \times p^x \times q^{(n-x)}$$

where $(n) = \frac{n!}{x!(n-x)!}$ (the "combination" formula)

Example 1:

A fair coin is tossed **6 times**. What is the probability of getting **exactly 4 heads**?

Identify the parameters:

- This is **Binomial** ✓
- $n = 6$ (fixed trials)
- $p = 0.5$ (probability of head)
- $q = 0.5$
- $x = 4$ (we want 4 heads)

Calculate:

$$P(X = 4) = \frac{(6)!}{(4)!(2)!} \times (0.5)^4 \times (0.5)^2$$

$$\frac{(6)!}{(4)!(2)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{(4 \times 3 \times 2 \times 1) \times (2 \times 1)} = 15$$

$$\begin{aligned} P(X = 4) &= 15 \times 0.0625 \times 0.25 \\ &= 15 \times 0.015625 \\ &= 0.234375 \text{ (or } 15/64) \end{aligned}$$

Visual representation:

6 coin tosses: H H H H T T
 ↑ ↑ ↑ ↑
 4 heads out of 6

Probability $\approx 23.44\%$

Example 2:

A biased die is rolled **5 times**. The probability of landing on '6' is **0.2**. What is the probability of getting **at least 4 sixes**?

Identify the parameters:

- $n = 5$
- $p = 0.2$
- $q = 0.8$
- "At least 4" means: $P(X \geq 4) = P(X = 4) + P(X = 5)$

Calculate $P(X = 4)$:

$$\begin{aligned} P(X = 4) &= \binom{5}{4} \times (0.2)^4 \times (0.8)^1 \\ &= 5 \times 0.0016 \times 0.8 \\ &= 0.0064 \end{aligned}$$

Calculate $P(X = 5)$:

$$\begin{aligned} P(X = 5) &= \binom{5}{5} \times (0.2)^5 \times (0.8)^0 \\ &= 1 \times 0.00032 \times 1 \\ &= 0.00032 \end{aligned}$$

Total:

$$P(X \geq 4) = 0.0064 + 0.00032 = 0.00672$$

Probability $\approx 0.67\%$

5.3 POISSON DISTRIBUTION [$X \sim \text{Po}(\lambda)$]

Context:

Used for **discrete data**, when we count the number of **events** happening in a **fixed interval** of time or space. Used for "**rare**" but randomly occurring events.

Conditions:

- ✓ Events occur at a **fixed, constant average rate** (denoted by λ , "lambda")
 - ✓ Events occur **independently**
 - ✓ Events **do not occur simultaneously**
-

Formula:

The probability of **x events** occurring in the interval is:

$$P(X = x) = (e^{-\lambda} \times \lambda^x) / x!$$

where λ = average rate

$e \approx 2.71828$ (a constant)

Example 3:

A hospital emergency room receives an average of **5 patients per hour**. What is the probability that it receives **exactly 2 patients** in a given hour?

Identify the parameters:

- This is **Poisson** ✓
- The interval is "per hour"
- Average rate: $\lambda = 5$
- We want: $x = 2$

Calculate:

$$\begin{aligned} P(X = 2) &= (e^{-5} \times 5^2) / 2! \\ &= (0.006738 \times 25) / 2 \\ &= 0.16845 / 2 \\ &= 0.0842 \end{aligned}$$

Probability $\approx 8.42\%$

Visual timeline:

Hour 1: |----•-----|
 Patient Patient
 1 2

Average: 5 patients/hour

Actual: 2 patients $\rightarrow P = 8.42\%$

Example 4:

On average, a website receives **60 hits per hour**. What is the probability it receives **2 hits in a 1-minute period**?

⚠ Important: Adjust λ to match the time interval!

Adjust λ :

$\lambda = 60 \text{ hits/hour} = 60 \text{ hits/60 minutes} = 1 \text{ hit/minute}$

Now use $\lambda = 1$ and $x = 2$:

Calculate:

$$\begin{aligned} P(X = 2) &= (e^{-1} \times 1^2) / 2! \\ &= (0.3679 \times 1) / 2 \\ &= 0.1839 \end{aligned}$$

Probability $\approx 18.39\%$

5.4 BINOMIAL APPROXIMATION BY POISSON

When to use:

Condition: If **n** is very large (e.g., $n > 50$) and **p** is very small (e.g., $p < 0.1$), the Binomial formula becomes **difficult to calculate**.

Approximation: We can use the **Poisson distribution** as an approximation, where:

$$\lambda = n \times p$$

Example 5:

A factory produces **1000 screws**. The probability that a screw is defective is **0.2%** (or $p = 0.002$). What is the probability that there are **exactly 3 defective screws** in the batch?

Binomial approach (Hard):

$$n = 1000, p = 0.002, x = 3$$

$$P(X = 3) = \frac{(1000) \times (0.002)^3 \times (0.998)^{997}}{(3)}$$

This is VERY difficult to calculate! ✗

Poisson approximation (Easy):

$$n = 1000 \text{ (large } \checkmark) \text{ and } p = 0.002 \text{ (small } \checkmark)$$

$$\lambda = n \times p = 1000 \times 0.002 = 2$$

$$\begin{aligned} P(X = 3) &= (e^{-2} \times 2^3) / 3! \\ &= (0.1353 \times 8) / 6 \\ &= 1.0824 / 6 \\ &= 0.1804 \end{aligned}$$

Probability $\approx 18.04\%$ ✓

The approximation is **very close** and **much faster**!

5.5 NORMAL DISTRIBUTION [$X \sim N(\mu, \sigma^2)$]

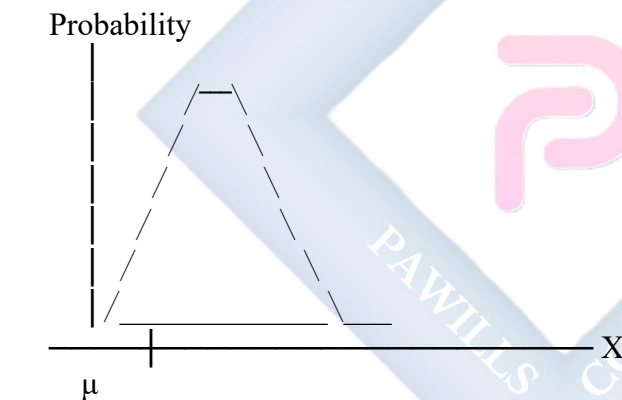
Context:

The **most important continuous distribution**. It describes any data that is **symmetric** and **"bell-shaped"** (e.g., heights, weights, test scores).

Parameters:

- μ (mu) = mean
- σ^2 (sigma squared) = variance
- σ (sigma) = standard deviation

The Bell Curve:



68% within $\mu \pm 1\sigma$
95% within $\mu \pm 2\sigma$
99.7% within $\mu \pm 3\sigma$

The Standard Normal Distribution (Z):

We cannot calculate probabilities for any normal curve directly. We must first **standardize** it, converting our X value into a **Z-score**.

A **Z-score** measures **how many standard deviations** an observation is from the mean.

Z-Score Formula:

$$Z = (X - \mu) / \sigma$$

The **Z-distribution** is a special normal distribution with $\mu = 0$ and $\sigma = 1$.

We use a **Standard Normal Table (Z-table)** to find probabilities for Z.

Using the Z-Table:

The table typically gives **P(Z < z)**, the area to the **left** of a Z-score.

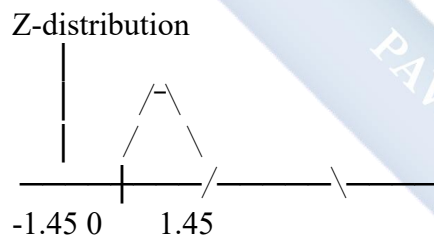
- $P(Z < 1.45)$ = Read directly from table (0.9265)

- $P(Z > 1.45)$ = $1 - P(Z < 1.45)$
= $1 - 0.9265$
= 0.0735

- $P(Z < -1.45)$ = (By symmetry) $P(Z > 1.45)$
= 0.0735

- $P(0.5 < Z < 1.5) = P(Z < 1.5) - P(Z < 0.5)$
= $0.9332 - 0.6915$
= 0.2417

Visual representation:



$$P(Z < -1.45) \quad P(Z > 1.45) \\ = 0.0735 \quad = 0.0735$$

Example 6:

WAEC scores are normally distributed with a **mean (μ) of 50** and a **standard deviation (σ) of 10**. [i.e., $X \sim N(50, 10^2)$]

What is the probability a student scores **above 65**?

Step 1: Find the Z-score

$$X = 65, \mu = 50, \sigma = 10$$

$$\begin{aligned} Z &= (X - \mu) / \sigma \\ &= (65 - 50) / 10 \\ &= 15 / 10 \\ &= 1.5 \end{aligned}$$

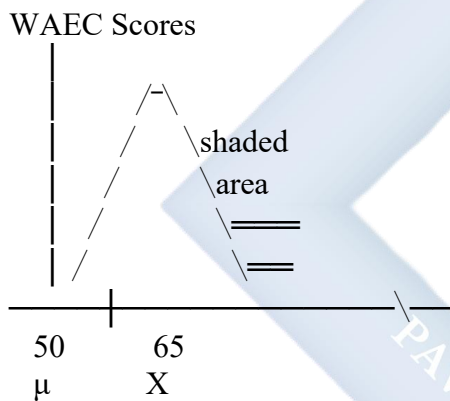
Step 2: Find the probability

$P(X > 65)$ is the same as $P(Z > 1.5)$

$$\begin{aligned} P(Z > 1.5) &= 1 - P(Z < 1.5) \\ &= 1 - 0.9332 \\ &= 0.0668 \end{aligned}$$

There is a 6.68% chance ✓

Visual representation:



$$P(X > 65) = 6.68\%$$

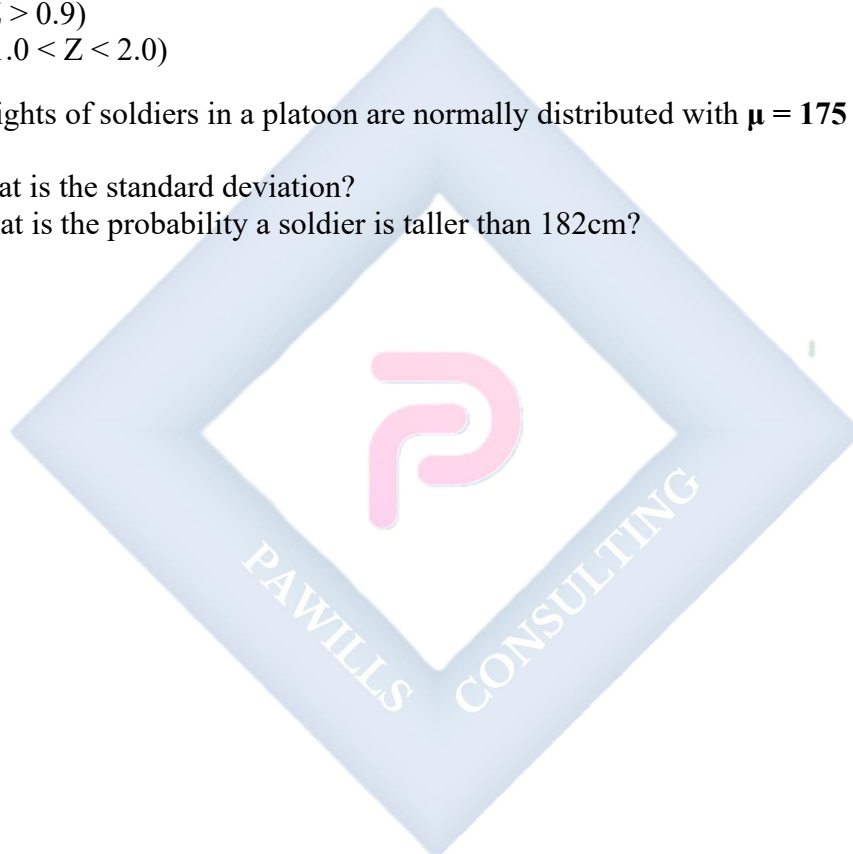
EVALUATION (WEEKS 5–6)

1. State the **4 conditions** for a Binomial distribution.
2. A student takes a **10-question** multiple-choice test (4 options per question). If they **guess every answer**, what is the probability they get **exactly 2 correct**?
3. A phone receives an average of **1.5 text messages per hour**. What is the probability it receives **0 text messages** in a given hour?
4. The weights of yams are normally distributed with a **mean of 3.5kg** and a **standard deviation of 0.5kg**. Find the probability that a randomly selected yam weighs **less than 3kg**.

5. State the **two conditions** under which a Binomial distribution can be approximated by a Poisson distribution.
-

ASSIGNMENT (WEEKS 5–6)

1. The probability of a light bulb being defective is **1%** ($p = 0.01$). In a box of **200 bulbs** ($n = 200$), what is the **(approximate) probability** of finding **exactly 4 defective bulbs**?
(Use the appropriate approximation)
2. Given that $Z \sim N(0, 1)$, find:
 - (a) $P(Z < -1.8)$
 - (b) $P(Z > 0.9)$
 - (c) $P(-1.0 < Z < 2.0)$
3. The heights of soldiers in a platoon are normally distributed with $\mu = 175 \text{ cm}$ and $\sigma^2 = 25 \text{ cm}^2$.
 - (a) What is the standard deviation?
 - (b) What is the probability a soldier is taller than 182cm?



WEEKS 7–8: VARIANCE

7.1 INTRODUCTION

Key Measures in Statistics:

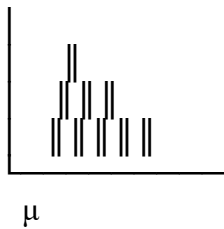
Mean (μ or $E[X]$):

A measure of **central tendency**. It is the "average" or "expected" value of a distribution.

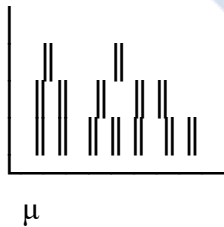
Variance (σ^2 or $\text{Var}(X)$):

A measure of **dispersion or spread**. It measures how far, on average, the data points are from the mean.

Low Variance → Data clustered tightly around mean (consistent)



High Variance → Data spread out (inconsistent)



Standard Deviation (σ):

The **square root** of the variance. It is used more often as it is in the **same units** as the mean.

$$\sigma = \sqrt{\text{Var}(X)} = \sqrt{\sigma^2}$$

Coefficient of Variation (CoV):

A **relative measure** of dispersion. It allows us to compare the variability of two different datasets, even if they have different means.

$$\text{CoV} = (\sigma/\mu) \times 100\%$$

This is useful when comparing datasets with different scales (e.g., comparing variability of chicken weights vs. cattle weights).

7.2 MEAN AND VARIANCE OF BINOMIAL DISTRIBUTION

For a Binomial distribution $X \sim B(n, p)$:

$$\text{Mean: } \mu = np$$

$$\text{Variance: } \sigma^2 = npq$$

$$\text{Std Dev: } \sigma = \sqrt{npq}$$

$$\text{where } q = 1 - p$$

Example 1:

A die is rolled **60 times**. Let X be the number of "**sixes**" obtained. Find the mean, variance, and coefficient of variation.

Identify parameters:

- This is **Binomial** ✓
- $n = 60$ (number of rolls)
- $p = 1/6$ (probability of rolling a six)
- $q = 5/6$ (probability of not rolling a six)

Calculate Mean:

$$\begin{aligned}\mu &= np \\ &= 60 \times (1/6) \\ &= 10\end{aligned}$$

We expect to get 10 sixes ✓

Calculate Variance:

$$\begin{aligned}\sigma^2 &= npq \\ &= 60 \times (1/6) \times (5/6) \\ &= 10 \times (5/6) \\ &= 50/6 \\ &\approx 8.33\end{aligned}$$

Calculate Standard Deviation:

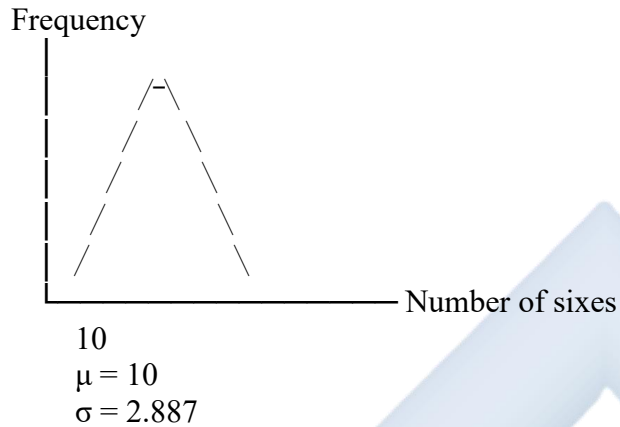
$$\begin{aligned}\sigma &= \sqrt{8.33} \\ &\approx 2.887\end{aligned}$$

Calculate Coefficient of Variation:

$$\begin{aligned}\text{CoV} &= (\sigma/\mu) \times 100\% \\ &= (2.887/10) \times 100\% \\ &= 28.87\%\end{aligned}$$

Visual representation:

Distribution of Sixes in 60 rolls:



Typical range: 7 to 13 sixes

7.3 MEAN AND VARIANCE OF POISSON DISTRIBUTION

For a Poisson distribution $X \sim \text{Po}(\lambda)$:

Mean:	$\mu = \lambda$
Variance:	$\sigma^2 = \lambda$
Std Dev:	$\sigma = \sqrt{\lambda}$

★ **Unique Property:** For Poisson distribution, **Mean = Variance = λ**

Example 2:

The number of accidents on a road follows a Poisson distribution with an average of **3.2 accidents per week**. Find the mean and standard deviation of the number of accidents.

Given:

- $\lambda = 3.2$

Calculate:

Mean: $\mu = \lambda = 3.2$ accidents/week

Variance: $\sigma^2 = \lambda = 3.2$

Std Dev: $\sigma = \sqrt{3.2} \approx 1.789$ accidents

Interpretation:

Week 1: ●●●○ (3 accidents)

Week 2: ●●●●○ (4 accidents)

Week 3: ●●○ (2 accidents)

Week 4: ●●●● (4 accidents)

Week 5: ●●● (3 accidents)

Average = 3.2 accidents/week

Typical variation = ± 1.789 accidents

7.4 MEAN AND VARIANCE OF NORMAL DISTRIBUTION

For a Normal distribution $X \sim N(\mu, \sigma^2)$:

The mean and variance are **given as the parameters** of the distribution.

Mean: μ
Variance: σ^2
Std Dev: σ

Example 3:

The weights of bags of rice are normally distributed, $X \sim N(50.1, 0.04)$.

Given:

$X \sim N(50.1, 0.04)$

Extract parameters:

Mean: $\mu = 50.1$ kg

Variance: $\sigma^2 = 0.04$ kg²

Std Dev: $\sigma = \sqrt{0.04} = 0.2$ kg

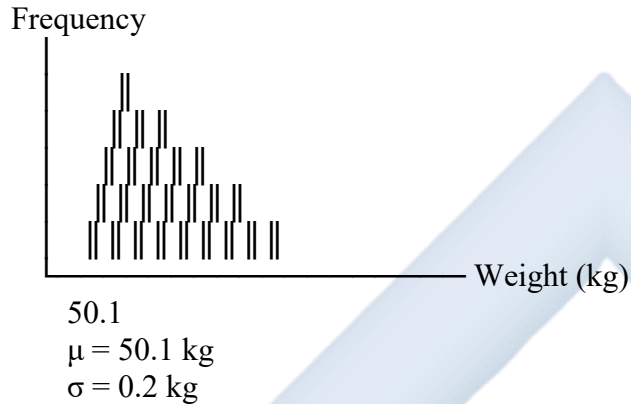
Calculate Coefficient of Variation:

$$\begin{aligned}\text{CoV} &= (\sigma/\mu) \times 100\% \\ &= (0.2/50.1) \times 100\% \\ &= 0.399\%\end{aligned}$$

Interpretation: The CoV is very low ($< 0.4\%$), which shows the weights are **very consistent** ✓

Visual representation:

Weight Distribution of Rice Bags:



Most bags: 49.9 kg to 50.3 kg
Very consistent! ✓

SUMMARY TABLE: Mean and Variance Formulas

Distribution	Mean (μ)	Variance (σ^2)	Std Dev (σ)
Binomial: $X \sim B(n,p)$	np	npq	$\sqrt{npq}$
Poisson: $X \sim \text{Po}(\lambda)$	λ	λ	$\sqrt{\lambda}$
Normal: $X \sim N(\mu, \sigma^2)$	μ	σ^2	σ

EVALUATION (WEEKS 7–8)

1. What does a **high variance** tell you about a dataset?
2. What is the main advantage of using the "**Coefficient of Variation**" instead of just the standard deviation?
3. A binomial distribution $X \sim B(100, 0.4)$. Find the mean, variance, and standard deviation.
4. The mean number of successes in a Poisson distribution is **9**. What is its standard deviation?
5. The scores on a test are $N(75, 16)$. State the mean and variance.

ASSIGNMENT (WEEKS 7–8)

1. A student fires **10 shots** at a target. The probability of hitting the target on any shot is **$p = 0.6$** . Let X be the number of hits.
 - (a) What distribution does X follow?
 - (b) Find the mean and standard deviation of X .
2. A textbook has **500 pages**. The number of misprints per page follows a Poisson distribution with a mean of **0.2**.
 - (a) What is the mean number of misprints in the entire 500-page book? (*Hint: Adjust λ*)
 - (b) What is the standard deviation of the number of misprints in the entire book?
3. Two datasets:
 - **Set A (Chicken Weights):** $\mu = 1.8$ kg, $\sigma = 0.6$ kg
 - **Set B (Cattle Weights):** $\mu = 150$ kg, $\sigma = 15$ kg
 - (a) Find the Coefficient of Variation for each set.
 - (b) Which set is relatively more variable?



WEEKS 9–10: CORRELATION

9.1 CONCEPT OF CORRELATION

Correlation is a statistical measure that describes the **strength** and **direction** of a **linear relationship** between two quantitative (numerical) variables.

Components of Correlation:

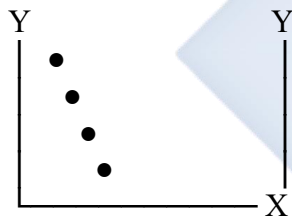
Strength: How closely the points follow a pattern

- **Strong:** Points are tightly clustered around a line
- **Weak:** Points are scattered
- **None:** No visible pattern

Direction:

POSITIVE CORRELATION (+)

As X ↑, Y ↑



Example: Height & Weight

NEGATIVE CORRELATION (-)

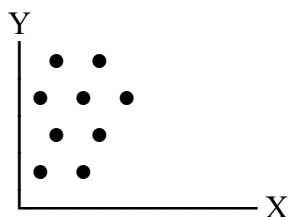
As X ↑, Y ↓



Example: TV hours & Exam Score

NO CORRELATION (0)

No relationship

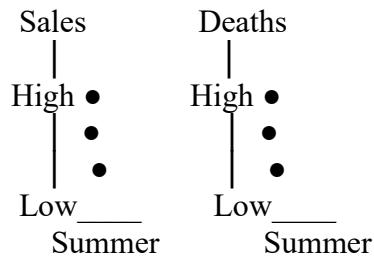


⚠ CRITICAL NOTE: Correlation $\neq$ Causation!

Just because two variables are correlated does **NOT** mean one causes the other.

Classic Example:

Ice Cream Sales $\leftrightarrow$ Drowning Deaths



They are correlated, but ice cream doesn't cause drowning!

Explanation: There is often a "lurking variable" (e.g., **hot weather**) that causes both.

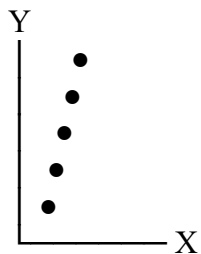
9.2 SCATTER DIAGRAMS

A **scatter diagram** (or scatter plot) is a graph used to visualize the relationship between two variables.

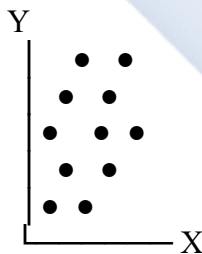
- Each pair of (x, y) values is plotted as a **single point**
- By looking at the "cloud" of points, we can instantly see the direction and strength of the correlation

Types of Scatter Plots:

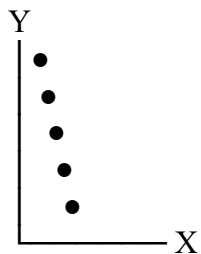
STRONG POSITIVE



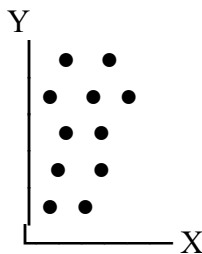
WEAK POSITIVE



STRONG NEGATIVE



NO CORRELATION



9.3 RANK CORRELATION (SPEARMAN'S RANK CORRELATION)

We often need to measure correlation when:

- The data is **not linear** (but still has a relationship, e.g., curved)
- The data is **ordinal** (can be ranked, but not measured precisely, e.g., "Best, Better, Good, Bad")

Spearman's Rank Correlation Coefficient (r_s) measures the strength of the relationship between the **ranks** of two variables.

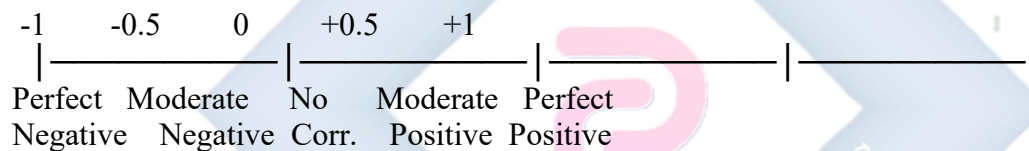
Properties of r_s :

r_s is always between -1 and +1

$r_s = +1 \rightarrow$ Perfect positive correlation (ranks are identical)

$r_s = -1 \rightarrow$ Perfect negative correlation (ranks are perfectly opposite)

$r_s = 0 \rightarrow$ No correlation



Formula (Simple, for No Ties):

$$r_s = \frac{1 - (6\sum d^2)}{n(n^2 - 1)}$$

where:

d = difference between the ranks for each pair

n = number of pairs

Example 1 (No Ties):

Two judges rank **5 competitors**. Find the Spearman's rank correlation.

Competitor	Judge A(Rank R_x)	Judge B(Rank R_y)	$d = R_x - R_y$	d^2
A	1	2	-1	1

Competitor	Judge A(Rank R_x)	Judge B(Rank R_y)	$d = R_x - R_y$	d^2
B	3	1	2	4
C	2	4	-2	4
D	5	5	0	0
E	4	3	1	1
n = 5			$\Sigma d^2 =$	10

Calculate:

$$\Sigma d^2 = 1 + 4 + 4 + 0 + 1 = 10$$

$$n = 5$$

$$n(n^2 - 1) = 5(25 - 1) = 5(24) = 120$$

$$\begin{aligned} r_s &= 1 - (6 \times 10)/120 \\ &= 1 - 60/120 \\ &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

✓ **Interpretation:** There is a **moderate positive correlation** between the judges' rankings ($r_s = 0.5$).

9.4 TIED RANKS

Rule: If two or more scores are tied, they must be given the **average of the ranks** they would have occupied.

Example of Tied Ranks:

Original Scores:

Scores: 10, 15, 15, 20, 25

Ranks: 1, ?, ?, 4, 5

↑ ↑
Tied for ranks 2 and 3

Calculate Average:

$$\text{Average} = (2 + 3)/2 = 2.5$$

Final Ranks:

Scores: 10, 15, 15, 20, 25

Ranks: 1, 2.5, 2.5, 4, 5 ✓

Example 2 (With Ties):

Find r_s for the following **Maths (X)** and **Physics (Y)** scores.

Step 1: Original Data

Student	Maths (X)	Physics (Y)
1	75	80
2	80	80
3	60	70
4	70	50
5	55	60
6	60	40

Step 2: Assign Ranks

Ranks for X (Maths):

Ordered: 80(1), 75(2), 70(3), 60(?), 60(?), 55(6)

↑ ↑
Tied for ranks 4 and 5
Average = $(4+5)/2 = 4.5$

Final Ranks: 80→1, 75→2, 70→3, 60→4.5, 60→4.5, 55→6

Ranks for Y (Physics):

Ordered: 80(?), 80(?), 70(3), 60(4), 50(5), 40(6)

↑ ↑
Tied for ranks 1 and 2
Average = $(1+2)/2 = 1.5$

Final Ranks: 80→1.5, 80→1.5, 70→3, 60→4, 50→5, 40→6

Step 3: Calculate Differences

Student	MathsRank (R_x)	PhysicsRank (R_y)	$d = R_x - R_y$	d^2
1	2	1.5	0.5	0.25
2	1	1.5	-0.5	0.25
3	4.5	3	1.5	2.25
4	3	5	-2	4

Student	MathsRank (R_x)	PhysicsRank (R_y)	$d = R_x - R_y$	d^2
5	6	4	2	4
6	4.5	6	-1.5	2.25
n = 6			$\Sigma d^2 =$	13

Step 4: Calculate r_s

$$\Sigma d^2 = 0.25 + 0.25 + 2.25 + 4 + 4 + 2.25 = 13$$

$$n = 6$$

$$n(n^2 - 1) = 6(36 - 1) = 6(35) = 210$$

$$\begin{aligned} r_s &= 1 - (6 \times 13)/210 \\ &= 1 - 78/210 \\ &= 1 - 0.371 \\ &= 0.629 \end{aligned}$$

✓ **Interpretation:** There is a **moderate-to-strong positive correlation** ($r_s = 0.629$) between Maths and Physics scores.

INTERPRETATION GUIDE FOR SPEARMAN'S r_s

Value of r_s	Interpretation	Visual
0.9 to 1.0	Very strong positive	●●●●●
0.7 to 0.9	Strong positive	●●●●○
0.5 to 0.7	Moderate positive	●●●○○
0.3 to 0.5	Weak positive	●●○○○
-0.3 to 0.3	Little or no correlation	●○○○○
-0.5 to -0.3	Weak negative	Similar patterns (negative)
-0.7 to -0.5	Moderate negative	
-0.9 to -0.7	Strong negative	
-1.0 to -0.9	Very strong negative	

EVALUATION (WEEKS 9–10)

1. Explain "**correlation does not imply causation**" using an example.
2. What is the main difference between **Pearson's correlation** and **Spearman's rank correlation**? (i.e., when would you use Spearman's?)
3. The scores for 5 students on a test are: **12, 18, 18, 10, 15**. Assign the correct ranks to these scores.

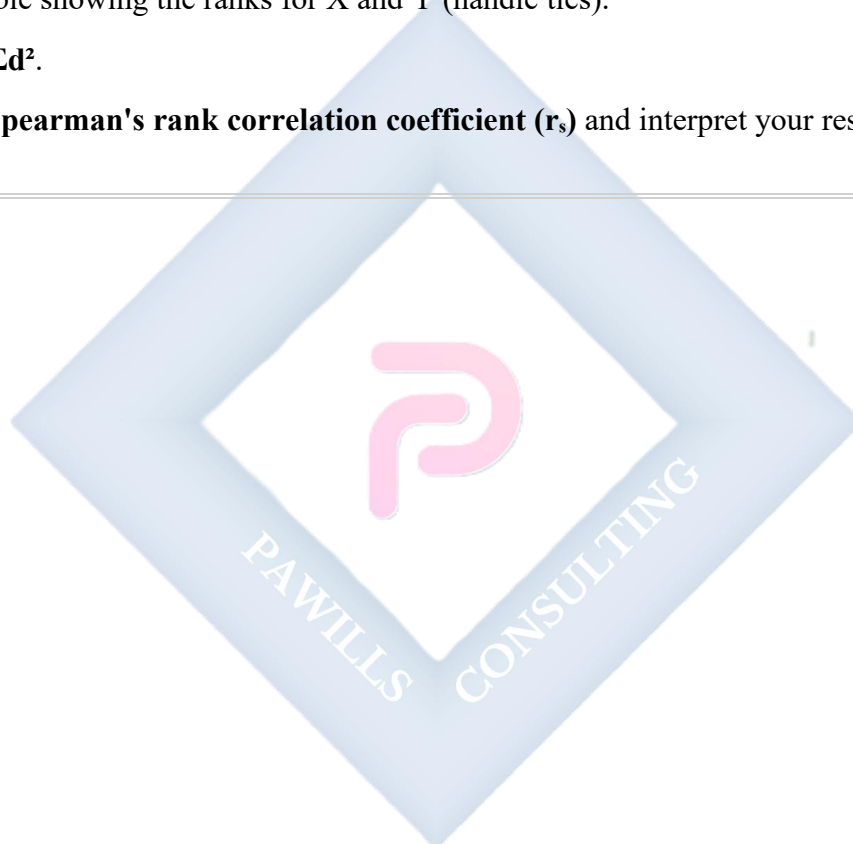
4. If $n = 7$ and $\Sigma d^2 = 28$, calculate Spearman's rank correlation coefficient.

ASSIGNMENT (WEEKS 9–10)

The following table shows the marks of **8 students** in Economics (X) and Government (Y).

Student	A	B	C	D	E	F	G	H
Econ (X)	30	45	60	38	50	45	70	30
Govt (Y)	55	60	70	50	65	58	80	60

- (a) Create a table showing the ranks for X and Y (handle ties).
(b) Calculate Σd^2 .
(c) Calculate **Spearman's rank correlation coefficient (r_s)** and interpret your result.
-



SS3 FURTHER MATHEMATICS LESSON NOTES

SECOND TERM

WEEKS 1–2: INTEGRATION

2.1 INTEGRATION AS THE REVERSE OF DIFFERENTIATION

Introduction:

Integration, or "anti-differentiation," is the **reverse process** of differentiation.

If we differentiate a function $f(x)$ to get $f'(x)$, then integrating $f'(x)$ brings us back to $f(x)$ (plus a constant).

The Constant Problem:

DIFFERENTIATION:

$$d/dx(x^3) = 3x^2$$

$$d/dx(x^3 + 5) = 3x^2$$

$$d/dx(x^3 - 10) = 3x^2$$

All give the same result! ↓

INTEGRATION:

$$\int 3x^2 dx = x^3 + C$$

We add C because we lost information about the original constant!

Visual representation:

Differentiation →

$$x^3 + 5 \longrightarrow 3x^2$$

$$x^3 + 0 \longrightarrow 3x^2$$

$$x^3 - 10 \longrightarrow 3x^2$$

← Integration

We can't tell which original function it came from!

So we write: $\int 3x^2 dx = x^3 + C$

The Indefinite Integral:

The symbol for integration is $\int$

$$\int f(x) dx = F(x) + C$$

where $F'(x) = f(x)$ and C = constant of integration

2.2 INTEGRATION OF ALGEBRAIC POLYNOMIALS

The Power Rule (for $n \neq -1$):

This is the **fundamental rule** for integrating polynomials.

$$\int ax^n dx = ax^{n+1}/(n+1) + C$$

In words:

"Add one to the power, then divide by the new power."

Memory aid:

Power goes UP by 1 $\uparrow$

Then DIVIDE by the new power $\downarrow$

$$\int x^2 dx = x^3/3 + C$$

$\uparrow \downarrow$
 $2+1$ divide by 3

Example 1: Find the integral of $f(x) = 4x^3 - 6x^2 + 8x - 5$

Solution:

Integrate **term by term**:

$$\begin{aligned} & \int (4x^3 - 6x^2 + 8x - 5) dx \\ &= 4x^{3+1}/(3+1) - 6x^{2+1}/(2+1) + 8x^{1+1}/(1+1) - 5x^{0+1}/(0+1) + C \\ &= 4x^4/4 - 6x^3/3 + 8x^2/2 - 5x/1 + C \\ &= x^4 - 2x^3 + 4x^2 - 5x + C \checkmark \end{aligned}$$

Step-by-step breakdown:

Term 1: $4x^3 \rightarrow 4x^4/4 = x^4$

Term 2: $-6x^2 \rightarrow -6x^3/3 = -2x^3$

Term 3: $8x \rightarrow 8x^2/2 = 4x^2$

Term 4: $-5 \rightarrow -5x/1 = -5x$

Constant: $+ C$

Integration of Functions of the Form $(ax + b)^n$:

$$\frac{\int (ax + b)^n dx = (ax + b)^{n+1}}{a(n + 1)} + C$$

Steps:

1. Treat the bracket as a single unit
2. Add one to the power
3. Divide by the new power
4. ALSO divide by the derivative of the inside (which is a)

Example 2: Integrate $\int (3x - 2)^4 dx$

Solution:

$$\int (3x - 2)^4 dx$$

$$= (3x - 2)^{4+1} / (3 \times (4+1)) + C$$

$$= (3x - 2)^5 / (3 \times 5) + C$$

$$= (3x - 2)^5 / 15 + C \checkmark$$

Breakdown:

Inside bracket: $(3x - 2)$

Derivative of inside: 3 (coefficient of x)

Power: $4 \rightarrow 5$

Denominator: $3 \times 5 = 15$

2.3 INTEGRATION INCLUDING LOGARITHMIC FUNCTIONS

The Problem with $n = -1$:

The power rule $\int x^n dx$ fails when $n = -1$ (i.e., $\int x^{-1} dx = \int 1/x dx$), as it would lead to **division by zero**.

$$\int x^n dx = x^{n+1}/(n+1) \leftarrow \text{Doesn't work when } n = -1! \\ \text{(gives } 0/0\text{)}$$

The Logarithmic Rule:

We know that $d/dx(\ln x) = 1/x$. Therefore:

$$\int 1/x dx = \ln|x| + C$$

We use $|x|$ (absolute value) because logarithm is only defined for positive numbers, while $1/x$ is defined for $x \neq 0$.

The General Logarithmic Rule (Pattern Recognition):

★ VERY IMPORTANT RULE:

If the **numerator** is the **derivative** of the **denominator**, the integral is the **natural log** of the denominator.

$$\int f'(x)/f(x) dx = \ln|f(x)| + C$$

Pattern to look for:

Derivative of bottom

↓

$$\int \frac{\text{Derivative of bottom}}{\text{bottom}} dx = \ln|\text{bottom}| + C$$

Example 3: Find $\int 2x/(x^2 + 1) dx$

Step 1: Identify the pattern

Let $f(x) = x^2 + 1$ (denominator)

Then $f'(x) = 2x$ (derivative)

Step 2: Check if numerator = derivative of denominator

Numerator = $2x$ ✓

$f'(x) = 2x$ ✓

Perfect match!

Step 3: Apply the logarithmic rule

$$\int 2x/(x^2 + 1) dx = \ln|x^2 + 1| + C \checkmark$$

Example 4: Find $\int x^2/(x^3 - 5) dx$

Step 1: Identify the function

Let $f(x) = x^3 - 5$ (denominator)

Then $f'(x) = 3x^2$ (derivative)

Step 2: Check the numerator

Numerator = x^2

But we need = $3x^2$ ✗

Not quite a match!

Step 3: "Force" the pattern

We can multiply by **3** and **divide by 3**:

$$\begin{aligned} \int x^2/(x^3 - 5) dx &= \int (1/3) \times (3x^2)/(x^3 - 5) dx \\ &= 1/3 \int (3x^2)/(x^3 - 5) dx \end{aligned}$$

Now it fits the pattern! ✓

$$= 1/3 \ln|x^3 - 5| + C \checkmark$$

Visual aid:

Original: $x^2/(x^3-5)$	Need: $3x^2/(x^3-5)$
↓	↓
Missing factor of 3	Add outside: $(1/3) \times 3$

2.4 DEFINITE INTEGRALS

A **definite integral** has **upper and lower limits** (bounds) and evaluates to a **single number**. It represents the **net area** under the curve between those bounds.

$$\begin{aligned} \int_a^b f(x) dx &= [F(x)]_a^b \\ &= F(b) - F(a) \end{aligned}$$

where:

$F(x)$ = anti-derivative of $f(x)$

a = lower limit
b = upper limit

Δ Important: The constant **C** is **NOT needed**, as it cancels out:

$$[F(b) + C] - [F(a) + C] = F(b) - F(a)$$

Example 5: Evaluate $\int_1^3 (3x^2 + 2) dx$

Step 1: Integrate

$$\begin{aligned}\int (3x^2 + 2) dx &= 3x^3/3 + 2x + C \\ &= x^3 + 2x + C\end{aligned}$$

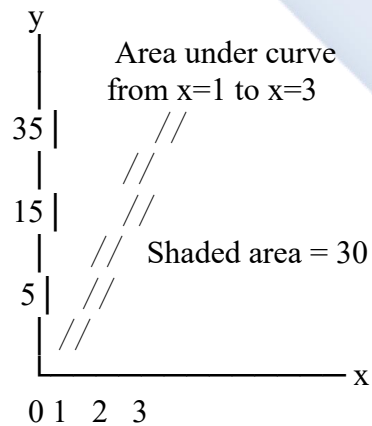
Step 2: Apply limits [1 to 3]

$$[x^3 + 2x]_1^3$$

Step 3: Evaluate (Upper limit - Lower limit)

$$\begin{aligned}&= [3^3 + 2(3)] - [1^3 + 2(1)] \\ &= [27 + 6] - [1 + 2] \\ &= 33 - 3 \\ &= 30 \checkmark\end{aligned}$$

Visual representation:



2.5 APPLICATION TO KINEMATICS

Kinematics is the study of motion. Integration is the tool we use to move from **acceleration** → **velocity** → **displacement**.

The Relationship:

DIFFERENTIATION $\longrightarrow$

Displacement $\rightarrow$ Velocity $\rightarrow$ Acceleration

$$\begin{array}{ccc} s(t) & v(t) & a(t) \\ & ds/dt & dv/dt = d^2s/dt^2 \end{array}$$

$\longleftarrow$ INTEGRATION

Integration Formulas:

$$v(t) = \int a(t) dt + C_1$$

$$s(t) = \int v(t) dt + C_2$$

The constants C_1 and C_2 are found using "initial conditions" (e.g., "at $t = 0$, $v = 5$ ")

Definite Integral Applications:

Change in velocity from t_1 to t_2 :

$$\Delta v = \int_{t_1}^{t_2} a(t) dt$$

Displacement from t_1 to t_2 :

$$\Delta s = \int_{t_1}^{t_2} v(t) dt$$

Δ Note: Total **distance** is $\int_{t_1}^{t_2} |v(t)| dt$ (different if velocity becomes negative!)

Example 6:

A particle starts **from rest** and moves with acceleration $a(t) = (12t - 4) \text{ m/s}^2$.

- (a) Find its velocity v at time t .
- (b) Find its displacement s at time t , if it starts at the origin.
- (c) Find the displacement from $t = 1$ to $t = 3$.

(a) Finding Velocity:

$$\begin{aligned} v(t) &= \int a(t) dt \\ &= \int (12t - 4) dt \end{aligned}$$

$$= 12t^2/2 - 4t + C_1$$

$$= 6t^2 - 4t + C_1$$

Apply initial condition: "Starts from rest" means $v = 0$ when $t = 0$

$$0 = 6(0)^2 - 4(0) + C_1$$

$$0 = C_1$$

Therefore:

$$v(t) = 6t^2 - 4t \text{ m/s } \checkmark$$

(b) Finding Displacement:

$$s(t) = \int v(t) \, dt$$

$$= \int (6t^2 - 4t) \, dt$$

$$= 6t^3/3 - 4t^2/2 + C_2$$

$$= 2t^3 - 2t^2 + C_2$$

Apply initial condition: "Starts at the origin" means $s = 0$ when $t = 0$

$$0 = 2(0)^3 - 2(0)^2 + C_2$$

$$0 = C_2$$

Therefore:

$$s(t) = 2t^3 - 2t^2 \text{ m } \checkmark$$

(c) Displacement from $t = 1$ to $t = 3$:

Using the definite integral:

$$\Delta s = \int_1^3 v(t) \, dt$$

$$= \int_1^3 (6t^2 - 4t) \, dt$$

$$= [2t^3 - 2t^2]_1^3$$

$$= [2(3)^3 - 2(3)^2] - [2(1)^3 - 2(1)^2]$$

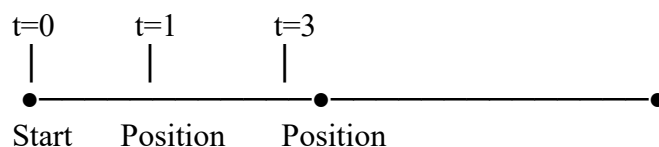
$$= [2(27) - 2(9)] - [2(1) - 2(1)]$$

$$= [54 - 18] - [2 - 2]$$

$$= 36 - 0$$

$$= 36 \text{ m } \checkmark$$

Visual timeline:



$$s=0 \quad s=0 \quad s=36\text{m}$$

Displacement from $t=1$ to $t=3 = 36\text{m}$

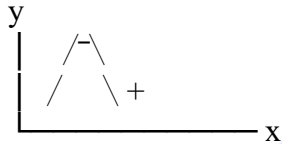
2.6 AREA UNDER A CURVE

The definite integral $\int_a^b f(x) \, dx$ gives the **net area** between:

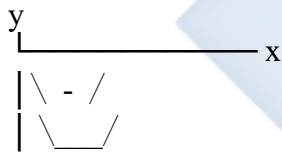
- The curve $y = f(x)$
- The **x-axis**
- The vertical lines $x = a$ and $x = b$

Sign Convention:

Area ABOVE x-axis $\rightarrow$ Positive integral



Area BELOW x-axis $\rightarrow$ Negative integral



If asked for "the area", take the absolute value of negative integrals!

Example 7:

Find the area enclosed by $y = x^2 - 5x + 4$, the x-axis, $x = 1$ and $x = 3$.

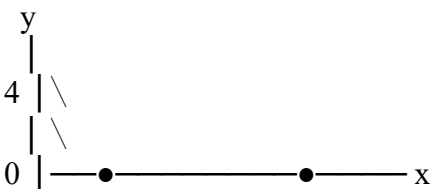
Step 1: Find the roots

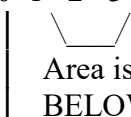
$$x^2 - 5x + 4 = 0$$

$$(x - 1)(x - 4) = 0$$

Roots: $x = 1$ and $x = 4$

Step 2: Sketch



0 1 2 3 4

 Area is
 BELOW x-axis

Between $x = 1$ and $x = 3$, the parabola is **below the x-axis**.

Step 3: Set up integral

$$\text{Area} = \int_1^3 (x^2 - 5x + 4) \, dx$$

Step 4: Integrate

$$= [x^3/3 - 5x^2/2 + 4x]_1^3$$

Step 5: Evaluate

$$\begin{aligned} &= [3^3/3 - 5(3^2)/2 + 4(3)] - [1^3/3 - 5(1^2)/2 + 4(1)] \\ &= [9 - 45/2 + 12] - [1/3 - 5/2 + 4] \\ &= [21 - 22.5] - [1/3 - 2.5 + 4] \\ &= -1.5 - 1.833... \\ &= -3.333... \\ &= -10/3 \end{aligned}$$

Step 6: Find area

Since area cannot be negative:

$$\text{Area} = |-10/3| = 10/3 \text{ square units } \checkmark$$

2.7 TRAPEZOIDAL RULE

This is a method of **numerical integration**, used to approximate a definite integral when:

- ✓ The function is too difficult or impossible to integrate (e.g., $\int e^{-x^2} \, dx$)
- ✓ We only have data points from an experiment, not an equation

The method approximates the area by dividing it into **n vertical strips** and treating each strip as a **trapezoid**.

The Formula:

Given **n strips** of equal width **h**, where **h = (b - a)/n**:

$$\int_a^b f(x) \, dx \approx h/2 [y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})]$$

$$y_{n-1})]$$

Where:

y_0 = first ordinate (at $x = a$)

y_n = last ordinate (at $x = b$)

$y_1, \dots, y_{n-1}$ = all "in-between" ordinates

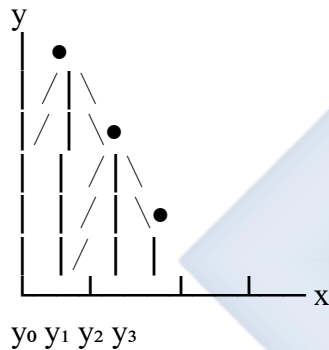
Memory aid:

FIRST + LAST + 2(ALL THE MIDDLE ONES)

$$y_0 + y_n + 2(y_1 + y_2 + \dots + y_{n-1})$$

Then multiply by $h/2$

Visual Representation:



Each strip is a trapezoid

Area of trapezoid = $h/2 \times (\text{sum of parallel sides})$

Example 8:

Estimate $\int_0^2 \frac{1}{(1+x^2)} dx$ using the trapezoidal rule with **4 strips**.

Step 1: Find h

$$a = 0, b = 2, n = 4$$

$$h = (b - a)/n = (2 - 0)/4 = 0.5$$

Step 2: Find ordinates

We need **5 ordinates** (y_0 to y_4) at $x = 0, 0.5, 1, 1.5, 2$

x	$y = 1/(1+x^2)$	Value
0	$y_0 = 1/(1+0^2)$	1.0000
0.5	$y_1 = 1/(1+0.5^2)$	0.8000

1.0		$y_2 = 1/(1 + 1^2)$		0.5000
1.5		$y_3 = 1/(1 + 1.5^2)$		0.3077
2.0		$y_4 = 1/(1 + 2^2)$		0.2000

Step 3: Apply formula

$$\text{Area} \approx h/2 [(y_0 + y_4) + 2(y_1 + y_2 + y_3)]$$

$$\approx 0.5/2 [(1.0 + 0.2) + 2(0.8 + 0.5 + 0.3077)]$$

$$\approx 0.25 [1.2 + 2(1.6077)]$$

$$\approx 0.25 [1.2 + 3.2154]$$

$$\approx 0.25 \times 4.4154$$

$$\approx 1.1039 \checkmark$$

2.8 VOLUME OF SOLIDS OF REVOLUTION

If a region bounded by a curve $y = f(x)$ is rotated 360° (2π radians) around an axis, it forms a **solid**. We can find its volume using integration.

Rotation about the x-axis:

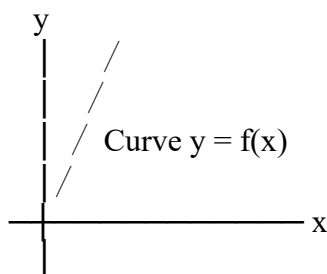
We sum an infinite number of thin cylinders ("discs") of **radius y** and **thickness dx**.

$$\text{Volume of one disc} = \pi r^2 \times h = \pi y^2 dx$$

Formula:

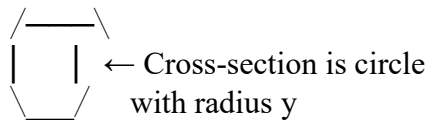
$$V_x = \pi \int_a^b y^2 dx$$

Visual representation:



Rotate 360° around x-axis ↻

Creates a solid:



Rotation about the y-axis:

We sum discs of **radius x** and **thickness dy**.

$$V_y = \pi \int_{cd} x^2 dy$$

(Note: You must rearrange the function to make x^2 the subject)

Example 9:

Find the volume of the solid formed when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 360° around the **x-axis**.

Step 1: Identify the formula

$$V_x = \pi \int_a^b y^2 dx$$

Step 2: Set up

$$a = 0, b = 4$$

$$y = \sqrt{x}$$

$$y^2 = x \checkmark$$

Step 3: Substitute

$$V_x = \pi \int_0^4 x dx$$

Step 4: Integrate

$$V_x = \pi [x^2/2]_0^4$$

Step 5: Evaluate

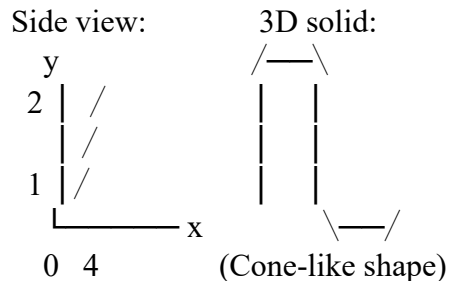
$$V_x = \pi [(4^2/2) - (0^2/2)]$$

$$= \pi [16/2 - 0]$$

$$= \pi \times 8$$

$$= 8\pi \text{ cubic units } \checkmark$$

Visual representation:



Volume = $8\pi \approx 25.13$ cubic units

EVALUATION (WEEKS 1–2)

- Find the indefinite integral of $f(x) = 12x^5 - 4/x^3 + 7$.
(Hint: $1/x^3 = x^{-3}$)
- Evaluate $\int_1^2 (6x^2 + 2)/(2x^3 + 2x) dx$.
- A particle's velocity is given by $v(t) = (3t^2 - 8t + 1)$ m/s. If its initial displacement is $s(0) = 5$ m, find its displacement at $t = 2$ s.
- Find the area enclosed by $y = x^2 + 2$, the x-axis, $x = 0$, and $x = 3$.
- Use the trapezoidal rule with **3 strips** to estimate $\int_2^5 x^2 dx$.
- Find the volume generated when the line $y = 2x$ is rotated about the x-axis between $x = 0$ and $x = 3$.

ASSIGNMENT (WEEKS 1–2)

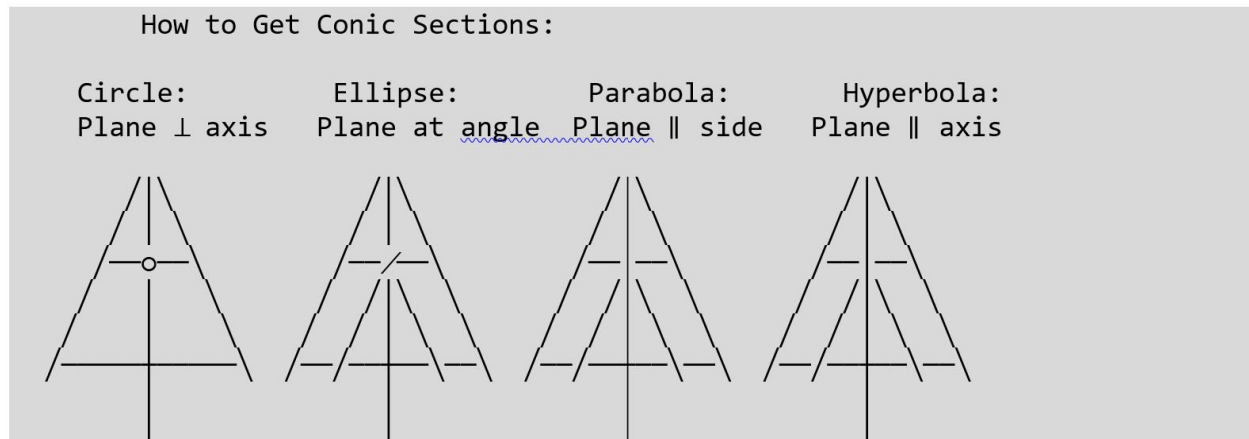
- Find $\int (4x - 1)^3 dx$.
- Find the area of the finite region bounded by the curve $y = x(x - 2)$ and the x-axis.
(Hint: Find the roots first to determine the limits a and b)
- A particle's acceleration is $a(t) = 10 \text{ m/s}^2$. Its velocity at $t = 1$ is **15 m/s**.
(a) Find an expression for its velocity $v(t)$.
(b) Find the total distance travelled from $t = 1$ to $t = 4$.
- The curve $y = 1/x$ is rotated about the x-axis between $x = 1$ and $x = 3$. Find the volume of the solid generated.

WEEKS 3–4: CONIC SECTIONS

3.1 INTRODUCTION

Conic sections are the curves obtained by intersecting a plane with a **double-napped cone**.

Visual Representation:



The Four Conics:

1. **Circle** - All points equidistant from center
2. **Parabola** - U-shaped curve
3. **Ellipse** - Oval shape
4. **Hyperbola** - Two separate curves

General Equation:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

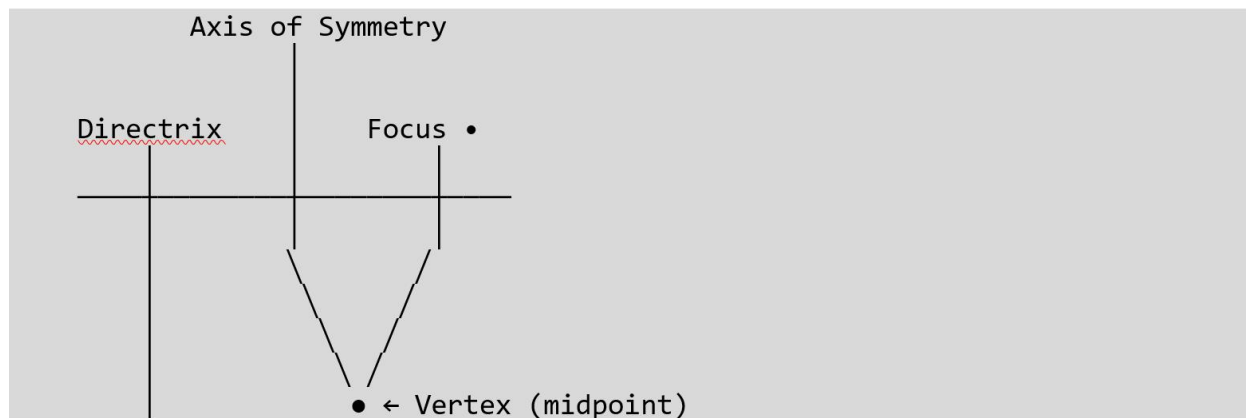
3.2 THE PARABOLA

Definition:

A **parabola** is the set of all points (x, y) in a plane that are **equidistant** from:

- A fixed point (the **focus**)
- A fixed line (the **directrix**)

Key Terms:



Vertex: The midpoint between the focus and directrix

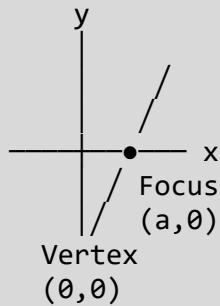
Axis of Symmetry: Line passing through vertex and focus

Standard Equations (Vertex at Origin):

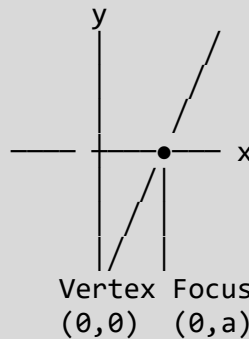
$y^2 = 4ax$	Opens RIGHT Focus: $(a, 0)$ Directrix: $x = -a$
$y^2 = -4ax$	Opens LEFT Focus: $(-a, 0)$ Directrix: $x = a$
$x^2 = 4ay$	Opens UPWARD Focus: $(0, a)$ Directrix: $y = -a$
$x^2 = -4ay$	Opens DOWNWARD Focus: $(0, -a)$ Directrix: $y = a$

Visual Guide:

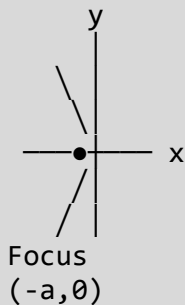
Opens RIGHT ($y^2 = 4ax$):



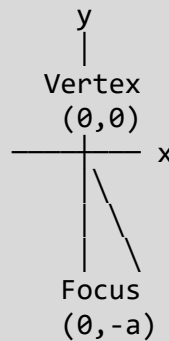
Opens UPWARD ($x^2 = 4ay$):



Opens LEFT ($y^2 = -4ax$):



Opens DOWNWARD ($x^2 = -4ay$):



Shifted Vertex (Vertex at (h, k)):

Opens RIGHT/LEFT: $(y - k)^2 = \pm 4a(x - h)$

Opens UP/DOWN: $(x - h)^2 = \pm 4a(y - k)$

Example 1: Find the focus and directrix of the parabola $y^2 = 12x$.

Step 1: Identify the form

This matches: $y^2 = 4ax$

Step 2: Find a

$$4a = 12$$

$$a = 3 \checkmark$$

Step 3: Determine opening direction

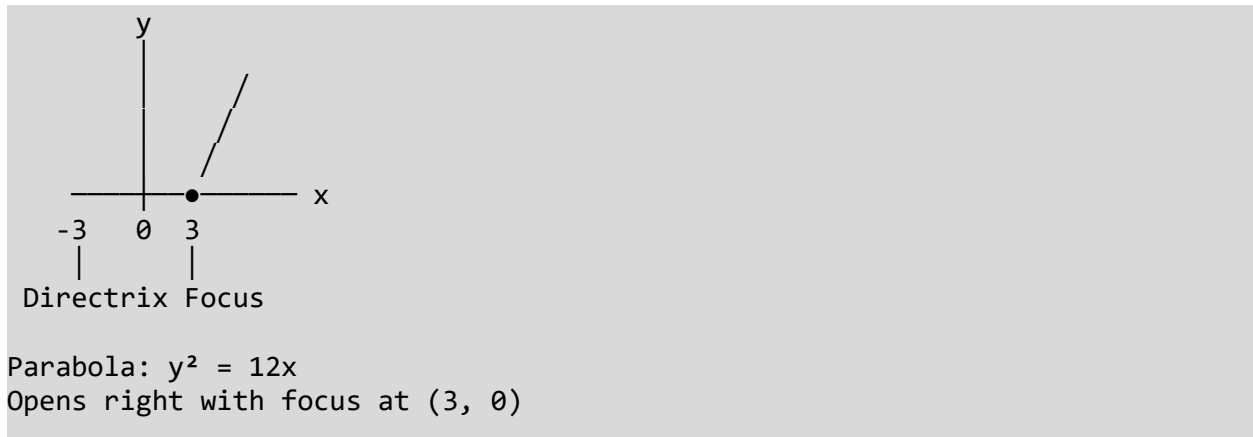
$$y^2 = 4ax \rightarrow \text{Opens to the RIGHT}$$

Step 4: Find focus and directrix

Focus: $(a, 0) = (3, 0) \checkmark$

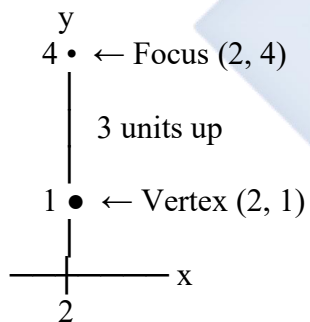
Directrix: $x = -a \rightarrow x = -3 \checkmark$

Visual representation:



Example 2: Find the equation of the parabola with vertex $(2, 1)$ and focus $(2, 4)$.

Step 1: Sketch the situation



Since focus is ABOVE vertex $\rightarrow$ Opens UPWARD

Step 2: Identify the form

Opens upward with shifted vertex:

$$(x - h)^2 = 4a(y - k)$$

Step 3: Find values

Vertex: $(h, k) = (2, 1)$

Distance from vertex to focus:

$$a = |4 - 1| = 3$$

Step 4: Write equation

$$(x - 2)^2 = 4(3)(y - 1)$$

$$(x - 2)^2 = 12(y - 1) \checkmark$$

Parametric Equations (for $y^2 = 4ax$):

A single parameter t is used to define both x and y :

$x = at^2$ $y = 2at$

Verification:

$$y^2 = (2at)^2 = 4a^2t^2$$

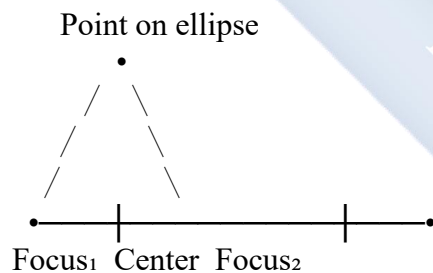
$$4ax = 4a(at^2) = 4a^2t^2$$

They match! $\checkmark$

3.3 THE ELLIPSE

Definition:

An **ellipse** is the set of all points (x, y) such that the **sum of the distances** from two fixed points (the **foci**) is a **constant** (which is **$2a$** , the length of the major axis).



$$\text{Distance}_1 + \text{Distance}_2 = \text{Constant } (2a)$$

Key Terms:

Eccentricity (e): A measure of how "non-circular" the ellipse is.

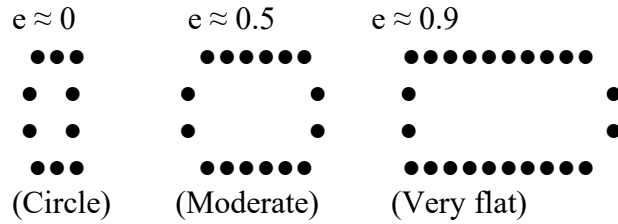
$$e = c/a$$

For an ellipse: $0 < e < 1$

$e = 0 \rightarrow$ Perfect circle

$e \rightarrow 1 \rightarrow$ Very long, thin ellipse

Visual representation of eccentricity:



Standard Equation (Centre at Origin):

$$x^2/a^2 + y^2/b^2 = 1$$

a = semi-major axis
 b = semi-minor axis
 where $a \geq b$

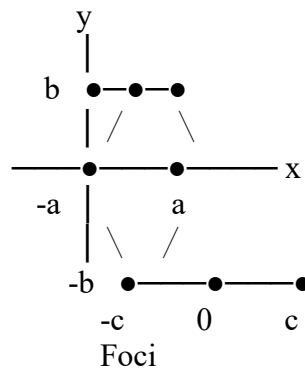
Foci Distance:

$$c^2 = a^2 - b^2$$

where c = distance from center to focus

Two Cases:

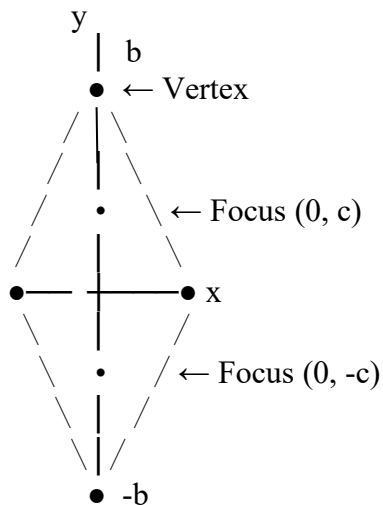
Case 1: Horizontal Major Axis (a^2 is under x^2)



Vertices: $(\pm a, 0)$

Foci: $(\pm c, 0)$

Case 2: Vertical Major Axis (b^2 is under y^2 and $b > a$)



Vertices: $(0, \pm b)$

Foci: $(0, \pm c)$

Example 3: Analyse the ellipse $x^2/25 + y^2/16 = 1$.

Step 1: Identify a and b

$$a^2 = 25 \rightarrow a = 5$$

$$b^2 = 16 \rightarrow b = 4$$

Step 2: Determine major axis

Since $a > b$:

Major axis is HORIZONTAL

Step 3: Find vertices

$$\text{Vertices: } (\pm a, 0) = (\pm 5, 0)$$

Points: $(-5, 0)$ and $(5, 0)$ ✓

Step 4: Find foci

$$c^2 = a^2 - b^2$$

$$c^2 = 25 - 16 = 9$$

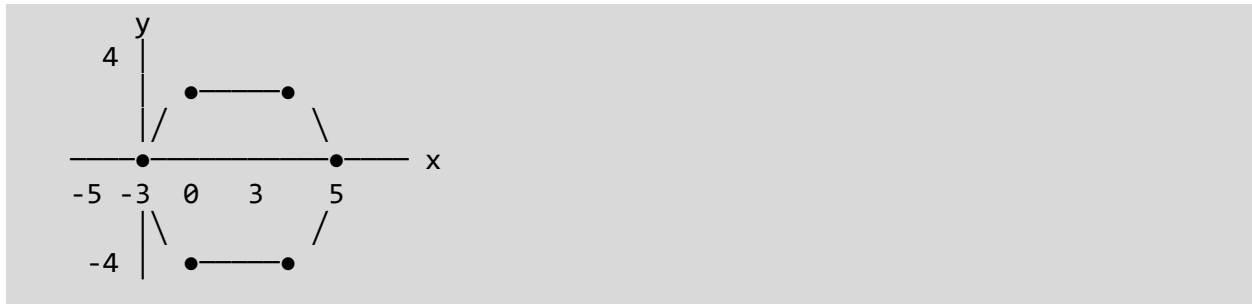
$$c = 3$$

$$\text{Foci: } (\pm c, 0) = (\pm 3, 0) \checkmark$$

Step 5: Find eccentricity

$$e = c/a = 3/5 = 0.6 \checkmark$$

Visual representation:



Foci at $(\pm 3, 0)$

Vertices at $(\pm 5, 0)$

$e = 0.6$ (moderately elliptical)

Parametric Equations:

$$x = a \cos \theta$$

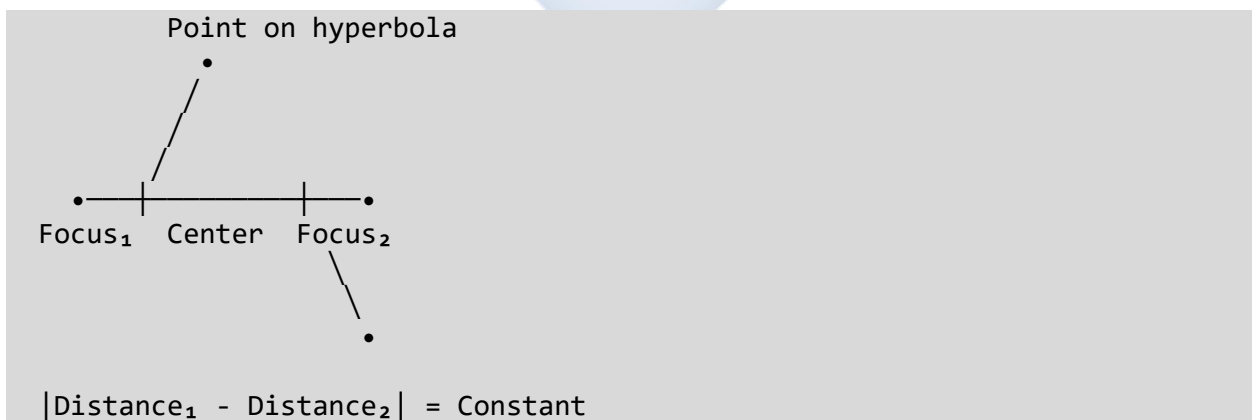
$$y = b \sin \theta$$

where θ is the
parameter (angle)

3.4 THE HYPERBOLA

Definition:

A **hyperbola** is the set of all points (x, y) such that the **difference of the distances** from two fixed points (the **foci**) is a **constant**.



Key Features:

- ✓ A hyperbola has **two branches**
- ✓ A hyperbola has **two asymptotes** (lines the curve approaches)
- ✓ **Eccentricity (e):** $e = c/a$, where $e > 1$ for hyperbolas

Standard Equations (Centre at Origin):

Horizontal Hyperbola:

$$x^2/a^2 - y^2/b^2 = 1$$

Opens: LEFT and RIGHT

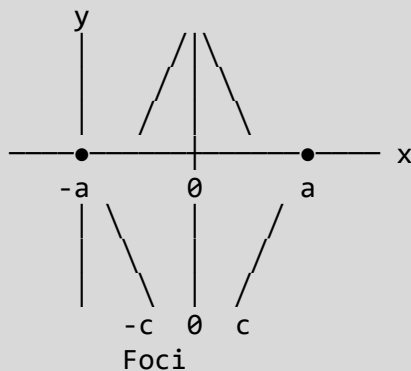
Vertices: $(\pm a, 0)$

Foci: $(\pm c, 0)$

$$c^2 = a^2 + b^2$$

Asymptotes: $y = \pm(b/a)x$

Visual:



Two branches opening horizontally
Asymptotes shown as dashed lines

Vertical Hyperbola:

$$y^2/a^2 - x^2/b^2 = 1$$

Opens: UP and DOWN

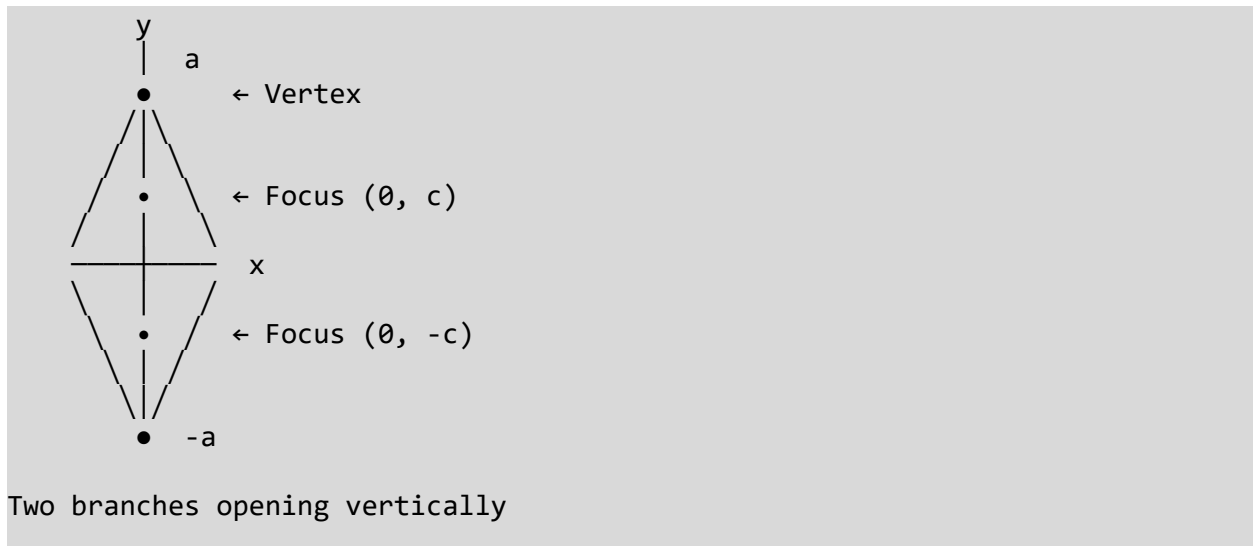
Vertices: $(0, \pm a)$

Foci: $(0, \pm c)$

$$c^2 = a^2 + b^2$$

Asymptotes: $y = \pm(a/b)x$

Visual:



Example 4: Analyse the hyperbola $9x^2 - 16y^2 = 144$.

Step 1: Convert to standard form

Divide by 144:

$$9x^2/144 - 16y^2/144 = 1$$

$$x^2/16 - y^2/9 = 1 \quad \checkmark$$

Step 2: Identify the type

$$\text{Form: } x^2/a^2 - y^2/b^2 = 1$$

This is a HORIZONTAL hyperbola $\checkmark$

Step 3: Find a and b

$$a^2 = 16 \rightarrow a = 4$$

$$b^2 = 9 \rightarrow b = 3$$

Step 4: Find vertices

$$\text{Vertices: } (\pm a, 0) = (\pm 4, 0) \quad \checkmark$$

Step 5: Find foci

$$c^2 = a^2 + b^2$$

$$c^2 = 16 + 9 = 25$$

$$c = 5$$

Foci: $(\pm c, 0) = (\pm 5, 0)$ ✓

Step 6: Find asymptotes

$$y = \pm(b/a)x$$

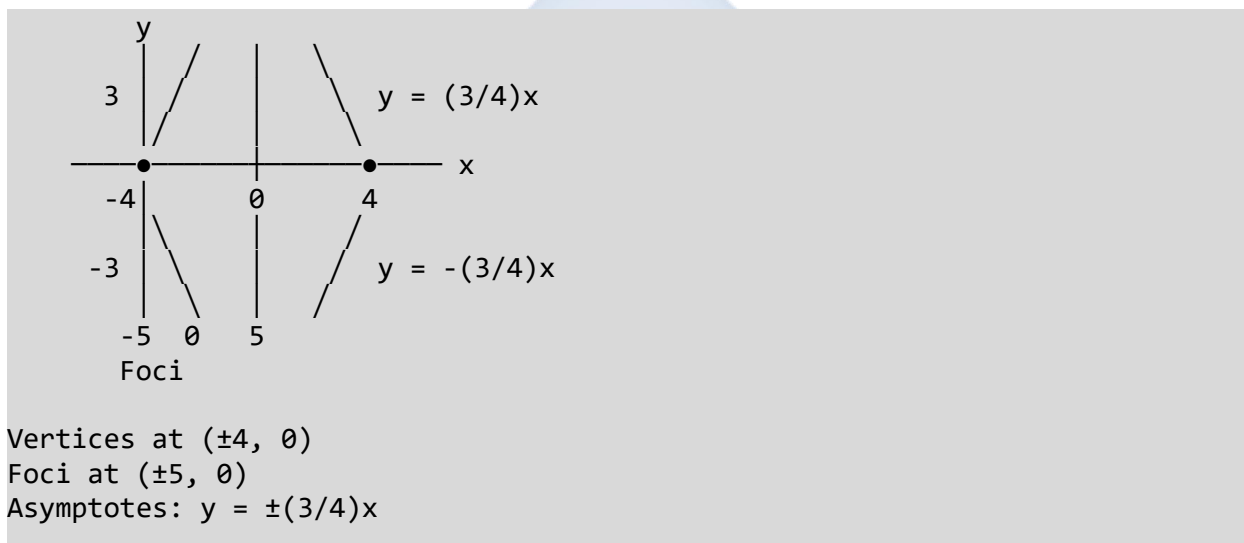
$$y = \pm(3/4)x \text{ ✓}$$

Step 7: Find eccentricity

$$e = c/a = 5/4 = 1.25 \text{ ✓}$$

Since $e > 1$, this confirms it's a hyperbola!

Visual representation:



Parametric Equations (Horizontal Hyperbola):

$$\begin{aligned} x &= a \sec \theta \\ y &= b \tan \theta \end{aligned}$$

SUMMARY TABLE: Conic Sections at a Glance

Conic	Standard Form	Key Features	Eccentricity
Circle	$x^2 + y^2 = r^2$	All points equidistant from center	$e = 0$
Parabola	$y^2 = 4ax$ or $x^2 = 4ay$	Equidistant from focus & directrix	$e = 1$
Ellipse	$x^2/a^2 + y^2/b^2 = 1$	Sum of distances = constant	$0 < e < 1$
Hyperbola	$x^2/a^2 - y^2/b^2 = 1$	Difference of distances = constant	$e > 1$

🔑 KEY FORMULAS REFERENCE:

Parabola:

$$y^2 = 4ax \rightarrow \text{Focus: } (a, 0), \text{ Directrix: } x = -a$$

$$x^2 = 4ay \rightarrow \text{Focus: } (0, a), \text{ Directrix: } y = -a$$

Ellipse:

$$x^2/a^2 + y^2/b^2 = 1 \rightarrow c^2 = a^2 - b^2$$

$$\text{Foci: } (\pm c, 0) \text{ or } (0, \pm c)$$

$$e = c/a$$

Hyperbola:

$$x^2/a^2 - y^2/b^2 = 1 \rightarrow c^2 = a^2 + b^2$$

$$\text{Foci: } (\pm c, 0) \text{ or } (0, \pm c)$$

$$\text{Asymptotes: } y = \pm(b/a)x \text{ or } y = \pm(a/b)x$$

$$e = c/a$$

EVALUATION (WEEKS 3–4)

1. Find the equation of a parabola with a **focus at (0, -5)** and a **directrix y = 5**.
 2. Find the **vertex, focus, and directrix** for the parabola $(y - 1)^2 = -8(x + 3)$.
 3. Find the **vertices, foci, and eccentricity** of the ellipse $25x^2 + 4y^2 = 100$.
 4. Find the **vertices, foci, and equations of the asymptotes** for the hyperbola $y^2/64 - x^2/36 = 1$.
 5. What conic section is represented by the parametric equations $x = 5\cos \theta$, $y = 2\sin \theta$? Find its Cartesian equation.
-

ASSIGNMENT (WEEKS 3–4)

1. A parabola has the equation $x^2 = 16y$.
(a) Find its focus and directrix.
(b) Find the parametric equations for this parabola.
2. Find the equation of the ellipse with **foci at $(\pm 8, 0)$** and **vertices at $(\pm 10, 0)$** .
3. Find the equation of the hyperbola with **vertices $(0, \pm 6)$** and **asymptotes $y = \pm 3x$** .

WEEKS 5–6: STATICS

5.1 FORCES IN EQUILIBRIUM

Statics is the branch of mechanics that deals with objects **at rest** or moving with **constant velocity** (i.e., **zero acceleration**).

Definition of Equilibrium:

A body is in **equilibrium** when it has:

1. **Zero resultant force** (no net force)
2. **Zero resultant moment** (no net turning effect)

Equilibrium = Balance!

No movement
OR
Constant velocity

Two Types of Equilibrium:

1. Translational Equilibrium:

The **vector sum** of all forces is zero.

$$\Sigma \vec{F} = 0$$

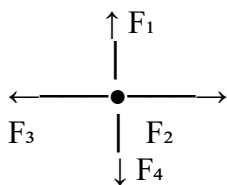
This implies:

$$\Sigma F_x = 0 \text{ (horizontal forces)}$$

$$\Sigma F_y = 0 \text{ (vertical forces)}$$

Visual representation:

All forces balance out:



$$F_1 + F_2 + F_3 + F_4 = 0^{\rightarrow}$$

2. Rotational Equilibrium:

The sum of all **moments** about any point is zero.

$$\Sigma \vec{M} = 0$$

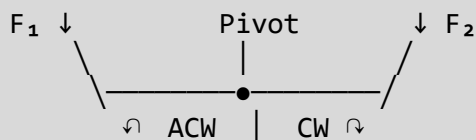
This implies:

Σ Clockwise Moments =

Σ Anti-Clockwise Moments

Visual representation:

Moments balance on a beam:



$$F_1 \times d_1 = F_2 \times d_2$$

(turning effects cancel)

5.2 FORCES AT A POINT

When several forces act at a **single point**, we only need to consider **translational equilibrium** ($\Sigma \vec{F} = 0$).

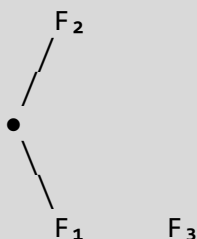
Methods for Solving:

Method 1: Triangle of Forces

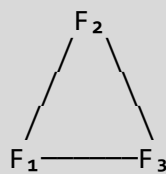
If **three forces** F_1 , F_2 , F_3 acting at a point are in equilibrium, they can be represented in **magnitude and direction** by the sides of a **closed triangle**.

Visual representation:

Forces at a point:



Triangle of forces:



The three forces form a closed triangle.

Method 2: Lami's Theorem

If **three forces** acting at a point are in equilibrium, then each force is proportional to the **sine of the angle** between the other two forces.

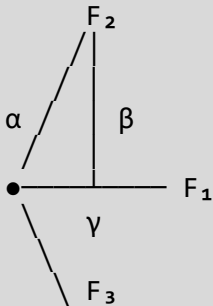
$$F_1/\sin \alpha = F_2/\sin \beta = F_3/\sin \gamma$$

where α = angle opposite F_1

β = angle opposite F_2

γ = angle opposite F_3

Visual representation:



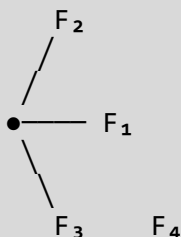
$$F_1/\sin \alpha = F_2/\sin \beta = F_3/\sin \gamma$$

Method 3: Polygon of Forces

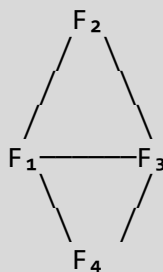
An extension of the triangle. If **n forces** acting at a point are in equilibrium, they can be represented by a **closed n-sided polygon**.

Visual representation:

4 forces at a point:



Closed quadrilateral:



Method 4: Resolution of Forces ★ (MOST POWERFUL METHOD)

This method works for **any number of forces**.

Steps:

1. **Draw a Free-Body Diagram (FBD)** showing all forces:
 - Weight (W)
 - Tension (T)
 - Normal Reaction (N)
 - Friction (f), etc.
2. **Choose axes:** horizontal (x-axis) and vertical (y-axis)
3. **Resolve every force** into x and y components:

A force F at angle θ to horizontal:

$$F_x = F \cos \theta$$

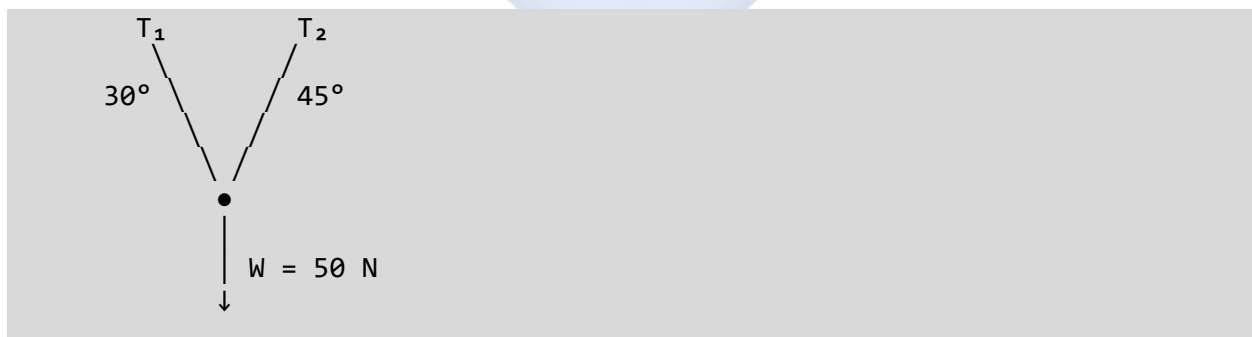
$$F_y = F \sin \theta$$

4. **Apply equilibrium conditions:**
 - $\Sigma F_x = 0$
 - $\Sigma F_y = 0$
5. **Solve the simultaneous equations**

Example 1:

A **5 kg mass** is suspended by two light, inelastic strings making angles of 30° and 45° with the horizontal. Find the tensions T_1 and T_2 in the strings. (Take $g = 10 \text{ m/s}^2$)

Step 1: Draw Free-Body Diagram



Step 2: Calculate Weight

$$W = mg = 5 \times 10 = 50 \text{ N (acts downwards)}$$

Step 3: Resolve Forces

T_1 components:

$$T_{1x} = -T_1 \cos 30^\circ \text{ (left, so negative)}$$

$$T_{1y} = +T_1 \sin 30^\circ \text{ (up, so positive)}$$

T_2 components:

$$T_{2x} = +T_2 \cos 45^\circ \text{ (right, so positive)}$$

$$T_{2y} = +T_2 \sin 45^\circ \text{ (up, so positive)}$$

W components:

$$W_x = 0$$

$$W_y = -50 \text{ (down, so negative)}$$

Step 4: Apply Equilibrium Conditions

Horizontal equilibrium ($\Sigma F_x = 0$):

$$-T_1 \cos 30^\circ + T_2 \cos 45^\circ = 0$$

$$T_2 \cos 45^\circ = T_1 \cos 30^\circ$$

$$T_2(0.7071) = T_1(0.866)$$

$$T_2 = 1.225T_1 \quad \dots (1)$$

Vertical equilibrium ($\Sigma F_y = 0$):

$$T_1 \sin 30^\circ + T_2 \sin 45^\circ - 50 = 0$$

$$T_1(0.5) + T_2(0.7071) = 50 \quad \dots (2)$$

Step 5: Solve Simultaneous Equations

Substitute (1) into (2):

$$0.5T_1 + (1.225T_1)(0.7071) = 50$$

$$0.5T_1 + 0.866T_1 = 50$$

$$1.366T_1 = 50$$

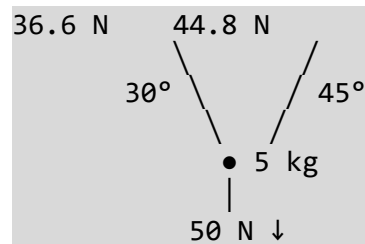
$$T_1 = 36.6 \text{ N } \checkmark$$

Find T_2 :

$$T_2 = 1.225 \times 36.6$$

$$T_2 = 44.8 \text{ N } \checkmark$$

Visual solution diagram:



System is in equilibrium $\checkmark$

5.3 MOMENT OF A FORCE & PARALLEL FORCES

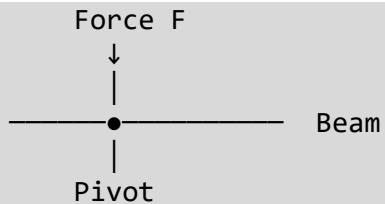
Moment of a Force:

The **turning effect** of a force about a pivot.

Moment = Force $\times$ Perpendicular
Distance from Pivot

$$M = F \times d$$

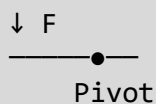
Visual representation:



d = perpendicular distance
Moment = $F \times d$

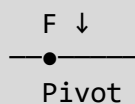
Direction of Moments:

CLOCKWISE (CW) ↻



Turns clockwise

ANTI-CLOCKWISE (ACW) ↺



Turns anti-clockwise

Principle of Moments:

For **rotational equilibrium**:

Σ Anti-Clockwise Moments =
 Σ Clockwise Moments

$$\Sigma \text{ ACWM} = \Sigma \text{ CWM}$$

Parallel Forces:

Forces that act **parallel** to each other.

Like Parallel Forces: (same direction)

$$\downarrow F_1 \quad \downarrow F_2$$

$$\text{Resultant: } R = F_1 + F_2 \downarrow$$

Unlike Parallel Forces: (opposite directions)

$$\downarrow F_1 \quad \uparrow F_2$$

$$\text{Resultant: } R = F_1 - F_2$$

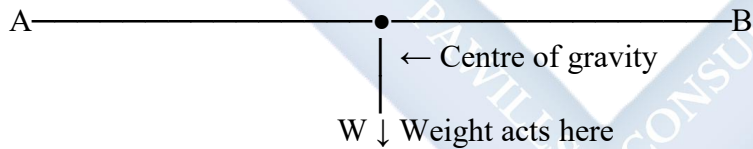
(direction of larger force)

Rigid Body:

A body that does **not deform** under force. For a **uniform rigid body** (like a beam), its weight **W** acts at its **centre of gravity** (geometric centre).

Visual representation:

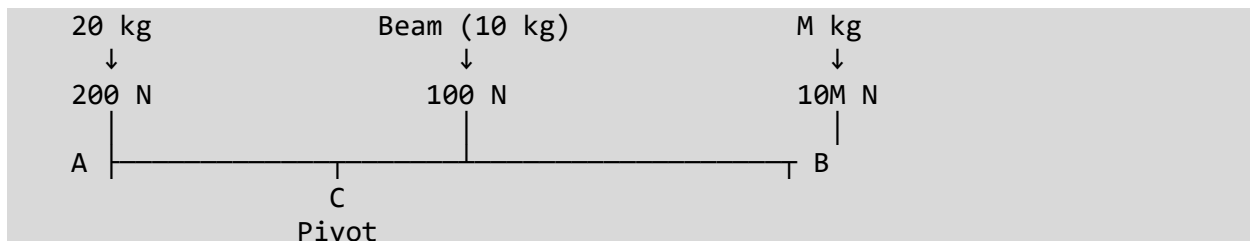
Uniform beam:

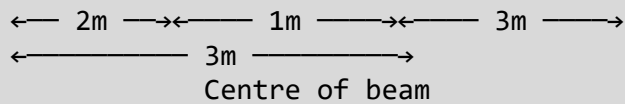


Example 2:

A uniform beam AB of length **6 m** and mass **10 kg** is pivoted at C, **2 m from A**. A **20 kg mass** hangs from A. What mass **M** must be hung at B for the beam to be in equilibrium? (Take $g = 10 \text{ m/s}^2$)

Step 1: Draw Free-Body Diagram





Step 2: Identify Forces and Distances

Force at A: $F_a = 20g = 200 \text{ N}$ (down)

Distance from C = 2 m (causes ACW moment $\curvearrowleft$)

Weight of beam: $W_b = 10g = 100 \text{ N}$ (down at centre)

Centre is at 3m from A

Distance from C = 3 - 2 = 1 m (causes CW moment $\curvearrowright$)

Force at B: $F_m = Mg = 10M \text{ N}$ (down)

Distance from C = 6 - 2 = 4 m (causes CW moment $\curvearrowright$)

Step 3: Apply Principle of Moments

Take moments about pivot C:

$$\Sigma \text{ ACWM} = \Sigma \text{ CWM}$$

$$F_a \times (\text{dist AC}) = W_b \times (\text{dist from C}) + F_m \times (\text{dist BC})$$

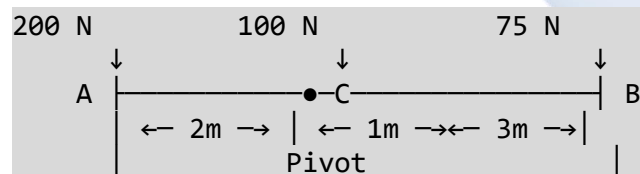
$$200 \times 2 = 100 \times 1 + 10M \times 4$$

$$400 = 100 + 40M$$

$$300 = 40M$$

$$M = 7.5 \text{ kg} \checkmark$$

Visual solution:



ACWM:

$$200 \times 2 = 400$$

CWM:

$$100 \times 1 + 75 \times 4 = 400$$

Balanced! $\checkmark$

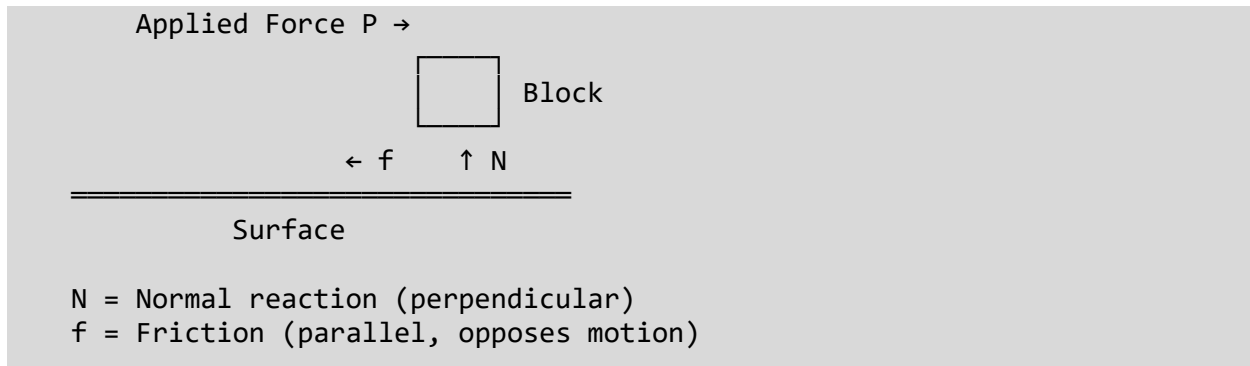
5.4 FRICTION

Definition:

Friction (f): A force that **opposes** relative motion or the tendency of motion between surfaces in contact. It acts **parallel** to the surface.

Normal Reaction (N): The force exerted by the surface **perpendicular** to itself, supporting the object.

Visual representation:



Types of Friction:

1. Static Friction (f_s):

Opposes the **start** of motion. It is a **variable force**:

$$0 \leq f_s \leq f_{\max}$$

2. Limiting Friction ($f_{\max}$):

The **maximum** value of static friction, occurring **just before** the object moves.

$$f_{\max} = \mu_s N$$

where μ_s = coefficient of static friction

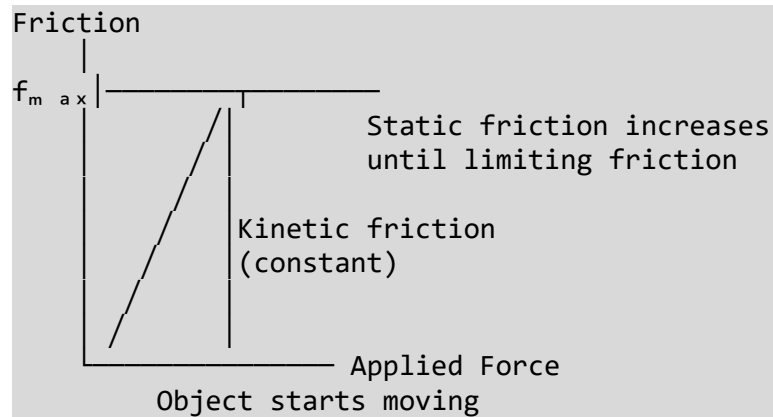
3. Kinetic Friction (f_k):

The friction force when the object is **already moving**.

$$f_k = \mu_k N$$

(Usually $\mu_k < \mu_s$)

Graph of Friction vs. Applied Force:



Angle of Friction (λ):

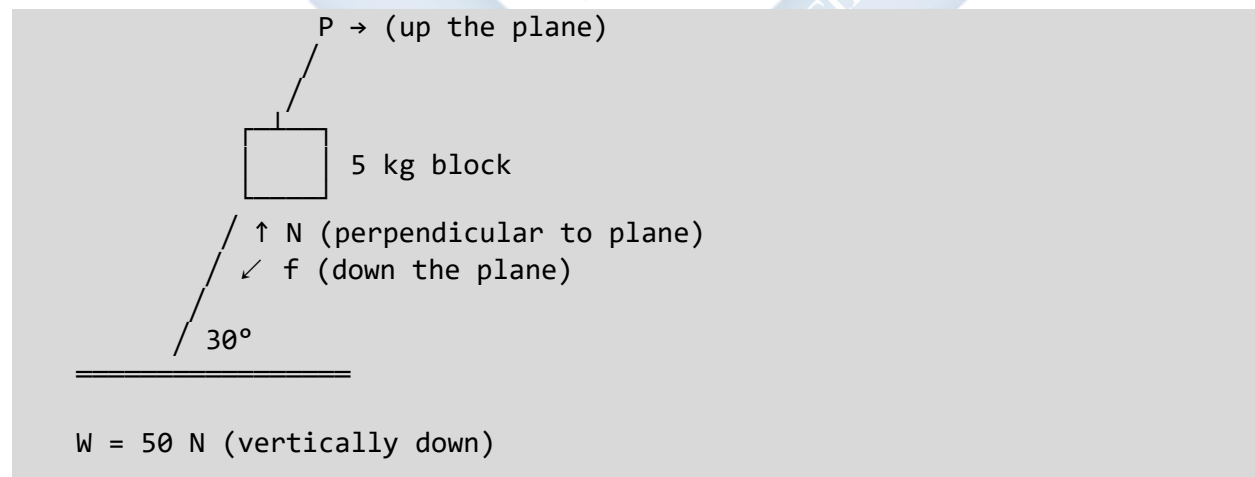
The angle such that:

$$\tan \lambda = \mu_s$$

Example 3: Block on a Rough Inclined Plane

A block of mass $m = 5 \text{ kg}$ rests on a rough plane inclined at $\theta = 30^\circ$. A force P pulls it up the plane. If $\mu_s = 0.4$, what is the **minimum force P** to start the block moving?

Step 1: Draw Free-Body Diagram



Step 2: Resolve Weight into Components

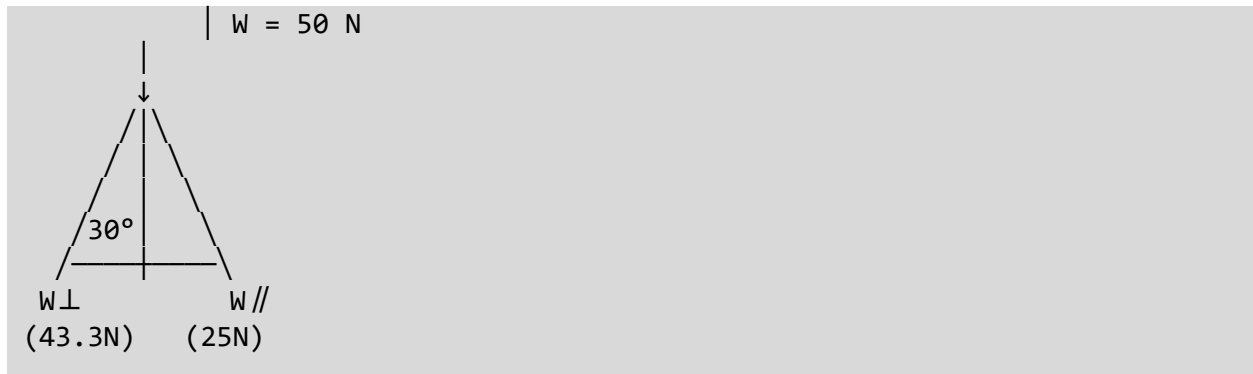
Parallel to plane: $W_{\parallel} = mg \sin 30^\circ$
 $= 50 \times 0.5$
 $= 25 \text{ N}$ (down plane)

Perpendicular to plane: $W_{\perp} = mg \cos 30^\circ$

$$= 50 \times 0.866$$

$$= 43.3 \text{ N (into plane)}$$

Visual resolution:



Step 3: Apply Equilibrium Conditions

Perpendicular to plane ($\Sigma F_{\perp} = 0$):

$$N - W_{\perp} = 0$$

$$N = W_{\perp} = 43.3 \text{ N} \checkmark$$

Parallel to plane ($\Sigma F_{\parallel} = 0$):

$$P \text{ (up)} - W_{\parallel} \text{ (down)} - f \text{ (down)} = 0$$

$$P - 25 - f = 0$$

$$P = 25 + f$$

Step 4: Find Minimum P

We want the **minimum P** to start motion. This occurs at **limiting friction**:

$$f = f_{\max} = \mu_s N$$

$$f_{\max} = 0.4 \times 43.3$$

$$f_{\max} = 17.32 \text{ N}$$

Therefore:

$$P_{\min} = 25 + f_{\max}$$

$$P_{\min} = 25 + 17.32$$

$$P_{\min} = 42.32 \text{ N} \checkmark$$

Visual summary:

Forces parallel to plane:

$$P = 42.32 \text{ N} \rightarrow \leftarrow f = 17.32 \text{ N} \leftarrow W_{\parallel} = 25 \text{ N}$$

$$42.32 = 25 + 17.32 \checkmark$$

Block is just about to move up!

SUMMARY TABLE: Key Friction Concepts

Type	Symbol	Formula	When it Acts
Static Friction	f_s	$0 \leq f_s \leq \mu_s N$	Object stationary
Limiting Friction	$f_{\max}$	$f_{\max} = \mu_s N$	Just before motion
Kinetic Friction	f_k	$f_k = \mu_k N$	Object moving

EVALUATION (WEEKS 5–6)

1. Define **translational** and **rotational** equilibrium.
 2. A **10 kg mass** is held by a force **P** acting horizontally and a string **T** at **30°** to the vertical. Find **P** and **T**.
 3. A uniform rod of length **4 m** and weight **50 N** is supported at its ends A and B. A **100 N weight** is placed **1 m from A**. Find the reaction forces at A and B.
 4. A **4 kg block** is on a horizontal surface. The coefficient of static friction is **0.5**. What is the maximum horizontal force that can be applied before the block moves? ($g = 10 \text{ m/s}^2$)
-

ASSIGNMENT (WEEKS 5–6)

1. Find the resultant of two forces, **8 N** and **5 N**, acting at a point with an angle of **60°** between them.
(Hint: Use the parallelogram law, $R^2 = P^2 + Q^2 + 2PQ \cos \theta$)
2. A **20 kg block** is on the verge of sliding down a rough inclined plane ($\mu_s = 0.25$). Find the **angle of inclination** of the plane.
(Hint: At this point, $f_{\max} (\text{up}) = W \parallel (\text{down})$)
3. Use **Lami's theorem** on Example 1 (the two strings) to find T_1 and T_2 . The angles between the forces are:
 - Between T_1 and W : $90^\circ + 30^\circ = 120^\circ$
 - Between T_2 and W : $90^\circ + 45^\circ = 135^\circ$
 - Between T_1 and T_2 : $180^\circ - 30^\circ - 45^\circ = 105^\circ$Verify that you get the same answer.

WEEK 7: MATHEMATICAL MODELLING

7.1 INTRODUCTION TO MATHEMATICAL MODELLING

What is Mathematical Modelling?

Mathematical modelling is the art and science of translating real-world problems into mathematical language—equations, graphs, functions, or inequalities. Think of it as building a bridge between reality and mathematics. Once we have this mathematical "model," we can solve it using mathematical tools, then translate the solution back to answer our original real-world question.

Important Note: A model is always a *simplification* of reality. We can't capture every detail, so we focus on the most important factors.

The Modelling Cycle

Mathematical modelling follows a circular process:

Step 1: Real-World Problem

Observe and identify a situation that needs investigation.

Example: "The fish population in Lake Chad is declining year after year."

Step 2: Make Assumptions & Identify Variables

Simplify the problem by making reasonable assumptions and defining clear variables.

Example:

- *Assumption:* The fish decline is primarily due to pollution (ignoring other factors like overfishing for now)
- *Variables:*
 - P = fish population
 - t = time (in years)
 - L = pollution level

Step 3: Construct the Model

Translate the problem into mathematical equations.

Example: "The rate at which the population decreases is proportional to the pollution level."

Mathematical form: $\frac{dP}{dt} = -kL$

where k is a constant of proportionality.

Step 4: Solve the Model

Use mathematical techniques (algebra, calculus, etc.) to find solutions.

Example: If pollution level L is constant, we can integrate: $P(t) = P_0 - kLt$

where P_0 is the initial population.

Step 5: Interpret & Validate

Translate your mathematical answer back into real-world language and check if it matches reality.

Example: "The fish population decreases linearly over time at a rate proportional to pollution level."

Validation: Compare predictions with actual data from the lake. If predictions don't match reality, refine your assumptions.

Refinement: "Perhaps fish also reproduce? Let's add a birth rate: $\frac{dP}{dt} = rP - kL$ "

7.2 DEPENDENT AND INDEPENDENT VARIABLES

Understanding the Relationship

In any model, we have two key types of variables:

Independent Variable (x)

- The **input** or **control** variable
- The variable that changes naturally or that you manipulate
- Often represents: time, price, age, dosage, temperature, etc.

Dependent Variable (y)

- The **output** or **response** variable
- Its value *depends on* the independent variable
- Often represents: population, profit, height, recovery rate, etc.

The Model Relationship: $y = f(x)$

The dependent variable is a function of the independent variable.

Practical Examples

Scenario	Independent Variable (x)	Dependent Variable (y)
Plant growth study	Time (days)	Height of plant (cm)
Drug effectiveness	Dosage (mg)	Patient recovery rate (%)

Scenario	Independent Variable (x)	Dependent Variable (y)
Business sales	Advertising spending (₹)	Monthly sales (₹)
Physics experiment	Temperature (°C)	Volume of gas (L)
Education study	Hours of study per week	Exam score (%)

Example Illustration: In a study on "Does exercise affect heart rate?", the researcher controls the *amount of exercise* (independent variable) and measures the resulting *heart rate* (dependent variable).

7.3 CONSTRUCTION OF MODELS & APPLICATIONS

Models range from simple linear equations to complex differential equations. Let's explore examples from different fields:

Application 1: Physical Science (Kinematics)

Problem: A student throws a ball straight up into the air with an initial velocity of 20 m/s. How high does it go?

Model Construction:

- **Variables:**
 - Independent: t = time (seconds)
 - Dependent: s = height (meters)
- **Assumptions:**
 - Gravity $g = 10 \text{ m/s}^2$ (constant)
 - Air resistance is negligible
 - The ball is a point mass

Mathematical Model: $s(t) = ut - \frac{1}{2}gt^2$

where u = initial velocity = 20 m/s

$$s(t) = 20t - 5t^2$$

Solution: Maximum height occurs when velocity = 0, i.e., when $\frac{ds}{dt} = 0$

$$\frac{ds}{dt} = 20 - 10t = 0 \Rightarrow t = 2 \text{ seconds}$$

$$s(2) = 20(2) - 5(2)^2 = 40 - 20 = 20 \text{ meters}$$

Interpretation: The ball reaches a maximum height of 20 meters after 2 seconds.

Application 2: Biological Science (Population Growth)

Problem: A bacteria colony in a petri dish doubles every hour. If we start with 100 bacteria, how many will there be after 5 hours?

Model Construction:

- **Variables:**
 - Independent: t = time (hours)
 - Dependent: P = population (number of bacteria)
- **Assumptions:**
 - Unlimited food and space
 - No bacteria die
 - Growth rate is proportional to current population

Mathematical Model (Malthusian Growth): $\frac{dP}{dt} = kP$

This is a differential equation stating "the rate of population increase is proportional to the current population."

Solution: $P(t) = P_0 e^{kt}$

where P_0 = initial population = 100

Since the population doubles every hour, when $t = 1$, $P = 200$: $200 = 100e^k \Rightarrow e^k = 2 \Rightarrow k = \ln(2) \approx 0.693$

Therefore: $P(t) = 100e^{0.693t}$

After 5 hours: $P(5) = 100e^{0.693 \times 5} = 100e^{3.465} \approx 100 \times 32 = 3,200$ bacteria

Interpretation: The bacteria population grows exponentially, reaching approximately 3,200 after 5 hours.

Limitation: This model assumes unlimited resources. In reality, growth slows as resources become scarce (leading to logistic growth models).

Application 3: Social/Economic Science (Business Profit)

Problem: A company manufactures notebooks. Each notebook sells for ₦500. The company has fixed costs (rent, salaries) of ₦50,000 per month, and each notebook costs ₦200 to produce. How many notebooks must be sold to break even? What's the profit if 300 notebooks are sold?

Model Construction:

- **Variables:**
 - Independent: x = number of notebooks sold
 - Dependent: P = profit (₦)

Mathematical Model:

Revenue: $R(x) = \text{price} \times \text{quantity} = 500x$

Cost: $C(x) = \text{Fixed Cost} + \text{Variable Cost } C(x) = 50,000 + 200x$

Profit: $P(x) = R(x) - C(x) \quad P(x) = 500x - (50,000 + 200x) \quad P(x) = 300x - 50,000$

This is a linear model of the form $y = mx + c$, where:

- $m = 300$ (profit margin per notebook)
- $c = -50,000$ (the starting deficit)

Solution:

Break-even point ($P = 0$): $0 = 300x - 50,000 \quad x = \frac{50,000}{300} \approx 167 \text{ notebooks}$

Profit for 300 notebooks: $P(300) = 300(300) - 50,000 = 90,000 - 50,000 = \text{₦}40,000$

Interpretation: The company must sell at least 167 notebooks to break even. Selling 300 notebooks yields ₦40,000 profit.

Illustrated Example: Taxi Fare Model

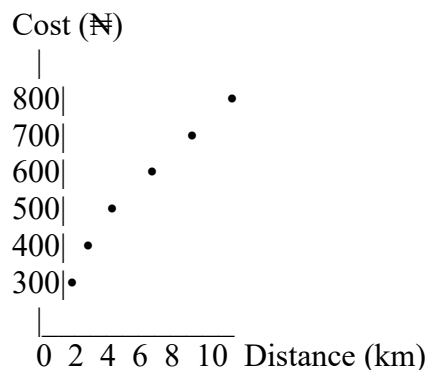
Problem: A taxi service charges ₦300 as a base fare plus ₦50 per kilometer traveled.

Model: $C(k) = 300 + 50k$

where:

- C = total cost (₦)
- k = distance traveled (km)

Graph:



- **Interpretation:** The y-intercept (₦300) is the fixed base fare. The slope (50) means each additional kilometer adds ₦50 to the cost.
-

EVALUATION (WEEK 7)

1. What is a mathematical model?

A mathematical model is a simplified mathematical representation of a real-world situation using equations, functions, graphs, or inequalities that allows us to analyze, predict, or understand the behavior of the system.

2. List the 5 main steps in the mathematical modelling cycle.

1. Identify the real-world problem
 2. Make assumptions and identify variables
 3. Construct the mathematical model
 4. Solve the model using mathematical techniques
 5. Interpret the solution and validate against reality
-

3. A company wants to model the "Effectiveness of Advertising" on "Sales".

(a) Identify the independent variable.

Answer: Effectiveness of Advertising (or Advertising Spending/Effort) is the independent variable, as it's what the company controls.

(b) Identify the dependent variable.

Answer: Sales is the dependent variable, as it depends on the advertising effort.

4. Write a simple mathematical model for the total cost C of a taxi ride, if there is a fixed "call-out" fee of ₦300 and a variable charge of ₦50 per kilometre k .

Answer: $C(k) = 300 + 50k$

where:

- C = total cost in naira
 - k = distance traveled in kilometers
-

ASSIGNMENT (WEEK 7)

1. Consider the population model $P(t) = P_0 e^{kt}$.

(a) What does P_0 represent?

Answer: P_0 represents the **initial population** at time $t = 0$. It's the starting point or baseline population before any growth or decline occurs.

(b) If $k > 0$, what does this imply about the population?

Answer: If $k > 0$, this implies **exponential growth**. The population is increasing over time at a rate proportional to its current size. The larger the value of k , the faster the growth.

(c) What is one major limitation of this model (i.e., an assumption that isn't realistic forever)?

Answer: The model assumes **unlimited resources and space**. In reality, every environment has a carrying capacity—a maximum population it can sustain. As resources become scarce, growth slows and eventually stops (described better by logistic growth models). Other limitations include: no consideration of deaths, diseases, predation, or migration.

2. A mobile phone company offers a plan: A fixed monthly charge of ₦2,000, which includes 100 free minutes. Any additional minute costs ₦10.

(a) Construct a mathematical model for the total monthly cost C as a function of the minutes used m . (This will be a "piecewise" function).

Answer:
$$C(m) = \begin{cases} 2000 & \text{if } 0 \leq m \leq 100 \\ 2000 + 10(m - 100) & \text{if } m > 100 \end{cases}$$

Simplified for $m > 100$:
$$C(m) = \begin{cases} 2000 & \text{if } 0 \leq m \leq 100 \\ 10m + 1000 & \text{if } m > 100 \end{cases}$$

(b) Use your model to calculate the cost for using 80 minutes.

Answer: Since $80 \leq 100$, we use the first part of the piecewise function: $C(80) = ₦2,000$

The customer pays only the fixed monthly charge.

(c) Use your model to calculate the cost for using 150 minutes.

Answer: Since $150 > 100$, we use the second part: $C(150) = 2000 + 10(150 - 100)$

$$C(150) = 2000 + 10(50) \quad C(150) = 2000 + 500 = ₦2,500$$

Alternatively: $C(150) = 10(150) + 1000 = 1500 + 1000 = ₦2,500$

WEEK 8: GAME THEORY

8.1 INTRODUCTION TO GAME THEORY

What is Game Theory?

Game Theory is the mathematical study of strategic decision-making in situations where the outcome depends not just on your own choices, but on the choices of others. It's like chess—your best move depends on what you think your opponent will do.

Key Terminology

Game: Any competitive or cooperative situation involving:

- **Players:** The decision-makers (people, companies, countries)
- **Strategies:** The complete plans of action available to each player
- **Payoffs:** The outcomes (rewards or losses) received after all players make their choices

Real-World Examples

- **Business:** Two companies deciding whether to lower prices
- **Politics:** Countries negotiating trade agreements
- **Sports:** A penalty kick in football (striker vs. goalkeeper)
- **Daily Life:** Choosing which route to take in traffic

8.2 TYPES OF GAMES

Classification of Games

1. Cooperative vs. Non-Cooperative

- *Cooperative:* Players can make binding agreements (e.g., business partnerships)
- *Non-Cooperative:* Players act independently (e.g., competing firms)

2. Simultaneous vs. Sequential

- *Simultaneous:* Players choose at the same time without knowing others' choices (e.g., rock-paper-scissors)
- *Sequential:* Players take turns (e.g., chess)

3. Zero-Sum vs. Non-Zero-Sum

- *Zero-Sum:* One player's gain equals another's loss (total payoff = 0)
- *Non-Zero-Sum:* All players can win or lose together

Focus: Two-Person Zero-Sum Games

We'll focus on games where:

- There are exactly 2 players (A and B)
- Whatever A wins, B loses (and vice versa)
- The total payoff is always zero

The Payoff Matrix

A **payoff matrix** shows Player A's payoff for every combination of strategies.

Example: Two friends, Ade and Bola, play a simple game:

	Bola: Left (B₁)	Bola: Right (B₂)
Ade: Up (A₁)	3	-1
Ade: Down (A₂)	0	2

Interpretation:

- If Ade chooses "Up" and Bola chooses "Left": Ade wins ₦3, Bola loses ₦3
- If Ade chooses "Up" and Bola chooses "Right": Ade loses ₦1, Bola wins ₦1
- If Ade chooses "Down" and Bola chooses "Left": Neither wins or loses (payoff = 0)
- If Ade chooses "Down" and Bola chooses "Right": Ade wins ₦2, Bola loses ₦2

Remember: The matrix always shows Player A's (row player's) payoff. Player B's payoff is the negative of this value.

8.3 SOLUTION OF TWO-PERSON ZERO-SUM GAMES

The Goals

- **Player A** (Row player): Wants to MAXIMIZE the payoff
- **Player B** (Column player): Wants to MINIMIZE the payoff (minimize A's gain = maximize B's gain)

Strategy 1: Pure Strategy (Finding a Saddle Point)

A **pure strategy** exists when there's a stable, predictable optimal choice for both players. We find this using the **Maximin-Minimax Principle**.

The Process

Step 1: Row Minima (Maximin for Player A)

- For each row, find the *minimum* payoff (worst case for A)
- Then find the *maximum* of these minimums
- This is A's "security level"—the most A can guarantee

Step 2: Column Maxima (Minimax for Player B)

- For each column, find the *maximum* payoff (worst case for B)
- Then find the *minimum* of these maximums
- This is B's "security level"—the least B can guarantee losing

Step 3: Check for Saddle Point

- If **Maximin = Minimax**, this value is a **saddle point**
- This is the **Value of the Game (V)**
- The optimal strategy is the row/column where this occurs

Example 1: Game WITH a Saddle Point

Problem: Find the optimal strategy for this game:

	B ₁	B ₂	B ₃	Row Min
A ₁	5	4	7	4
A ₂	-1	2	3	-1
A ₃	6	5	10	5 ←
Col Max	6	5 ↑	10	

Solution Process:

1. **Row Minima:** {4, -1, 5}
 - **Maximin** = $\max\{4, -1, 5\} = 5$ (A should play A₃)
2. **Column Maxima:** {6, 5, 10}
 - **Minimax** = $\min\{6, 5, 10\} = 5$ (B should play B₂)
3. **Saddle Point:** Maximin = Minimax = 5 ✓

Optimal Strategy:

- Player A plays strategy A₃
- Player B plays strategy B₂
- **Value of the Game:** $V = 5$

Interpretation: If both players play optimally, A will consistently win ₦5 per game. This game favors Player A.

Example 2: Game WITHOUT a Saddle Point

Problem: Analyze this game:

	B ₁	B ₂	B ₃	Row Min
A ₁	9	8	4	4 ←
A ₂	2	3	5	2

	B ₁	B ₂	B ₃	Row Min
A ₃	7	1	6	1
Col Max	9	8	6 ↑	

Solution Process:

1. **Row Minima:** {4, 2, 1}
 - **Maximin** = $\max\{4, 2, 1\} = 4$
2. **Column Maxima:** {9, 8, 6}
 - **Minimax** = $\min\{9, 8, 6\} = 6$
3. **Check:** Maximin (4) $\neq$ Minimax (6) ✗

Conclusion: No saddle point exists. This game requires a **mixed strategy** (randomization).

Strategy 2: Mixed Strategy (for 2×2 Games)

When no saddle point exists, players must **randomize** their strategies to remain unpredictable.

The Setup

For a 2×2 game:

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

- **Player A** plays A₁ with probability p_1 and A₂ with probability p_2 (where $p_1 + p_2 = 1$)
- **Player B** plays B₁ with probability q_1 and B₂ with probability q_2 (where $q_1 + q_2 = 1$)

The Formulas (Odds Method)

For Player A's probabilities:

1. Find row differences:
 - $d_1 = |a - b|$
 - $d_2 = |c - d|$
2. Calculate probabilities:

$$p_1 = \frac{d_2}{d_1 + d_2}$$

$$p_2 = \frac{d_1}{d_1 + d_2}$$

For Player B's probabilities:

1. Find column differences:
 - $d^c_1 = |a - c|$
 - $d^c_2 = |b - d|$

2. Calculate probabilities:

$$q_1 = \frac{d^{c_2}}{d^{c_1} + d^{c_2}}$$
$$q_2 = \frac{d^{c_1}}{d^{c_1} + d^{c_2}}$$

Value of the Game: $V = \frac{ad-bc}{a+d-(b+c)}$

Example 3: Solving a Mixed Strategy Game

Problem: Two players play the following game. Find the optimal mixed strategies.

	B ₁	B ₂	Row Min
A ₁	4	1	1
A ₂	2	3	2
Col Max	4	3	

Check for Saddle Point:

- Maximin = 2
- Minimax = 3
- Since $2 \neq 3$, no saddle point exists → Use mixed strategy

Solution:

Matrix: $M = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$

So: a = 4, b = 1, c = 2, d = 3

Step 1: Find Player A's Strategy

Row differences:

- $d_1 = |4 - 1| = |3| = 3$
- $d_2 = |2 - 3| = |-1| = 1$

Probabilities: $p_1 = \frac{d_2}{d_1 + d_2} = \frac{1}{3+1} = \frac{1}{4} = 0.25$

$$p_2 = \frac{d_1}{d_1 + d_2} = \frac{3}{3+1} = \frac{3}{4} = 0.75$$

Player A's Optimal Strategy: Play A₁ 25% of the time, play A₂ 75% of the time.

Step 2: Find Player B's Strategy

Column differences:

- $d^c_1 = |4 - 2| = |2| = 2$
- $d^c_2 = |1 - 3| = |-2| = 2$

Probabilities: $q_1 = \frac{d^c_2}{d^c_1 + d^c_2} = \frac{2}{2+2} = \frac{2}{4} = 0.5$

$$q_2 = \frac{d^c_1}{d^c_1 + d^c_2} = \frac{2}{2+2} = \frac{2}{4} = 0.5$$

Player B's Optimal Strategy: Play B₁ 50% of the time, play B₂ 50% of the time.

Step 3: Find the Value of the Game

$$V = \frac{ad - bc}{a + d - (b + c)}$$
$$V = \frac{(4)(3) - (1)(2)}{4 + 3 - (1 + 2)}$$
$$V = \frac{12 - 2}{7 - 3} = \frac{10}{4} = 2.5$$

Interpretation:

- If both players follow their optimal mixed strategies, Player A will win an average of ₦2.50 per game in the long run
- This game favors Player A
- Neither player can improve their expected payoff by deviating from this strategy if the opponent plays optimally

Practical Example: Business Competition

Scenario: Two mobile phone companies (MTN and Glo) are deciding whether to run a promotion. The payoffs (in millions of naira market share gain for MTN) are:

	Glo: Promote	Glo: Don't Promote
MTN: Promote	-1	3
MTN: Don't Promote	-2	1

Analysis:

- If both promote: MTN loses ₦1M market share (intense competition)
- If only MTN promotes: MTN gains ₦3M (attracts customers)

- If only Glo promotes: MTN loses ₦2M (customers leave)
- If neither promotes: MTN gains ₦1M (natural advantage)

Check for Saddle Point:

- Row minima: $\{-1, -2\} \rightarrow \text{Maximin} = -1$
- Column maxima: $\{-1, 3\} \rightarrow \text{Minimax} = -1$
- Saddle point exists at $(-1, -1)$!

Optimal Strategy: Both companies should promote. MTN loses ₦1M, but this is the "least worst" stable outcome.

EVALUATION (WEEK 8)

1. What is a "zero-sum game"?

A zero-sum game is a competitive situation where the total payoff to all players sums to zero. Whatever one player wins, the other player(s) lose by exactly the same amount. One player's gain is precisely another player's loss. Examples include poker, futures contracts, and strictly competitive sports.

2. Explain the "Maximin-Minimax" principle.

The **Maximin-Minimax principle** is a method for finding optimal pure strategies:

- **Maximin (Player A's perspective):** Find the minimum payoff in each row (worst-case scenario), then choose the row with the maximum of these minimums. This guarantees Player A the best worst-case outcome.
- **Minimax (Player B's perspective):** Find the maximum payoff in each column (worst-case scenario for B), then choose the column with the minimum of these maximums. This guarantees Player B limits their maximum loss.

If $\text{Maximin} = \text{Minimax}$, a saddle point exists and both players have a clear optimal pure strategy.

3. What is a "saddle point" and what does it tell us?

A **saddle point** is a value in the payoff matrix where:

- It is the minimum value in its row (worst for A in that row)
- It is the maximum value in its column (worst for B in that column)
- $\text{Maximin} = \text{Minimax}$ at this point

What it tells us:

- Both players have optimal **pure strategies** (no randomization needed)

- This is the **value of the game** (expected payoff if both play optimally)
- The game is **stable**—neither player benefits from changing strategy if the opponent plays optimally
- It represents the **equilibrium** of the game

4. Find the saddle point (if any) and the value of the game for the matrix:

$$\begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix}$$

Solution:

	B ₁	B ₂	Row Min
A ₁	1	0	0 ←
A ₂	-2	3	-2
Col Max	1	3 ↑	

- Row minima: {0, -2} → **Maximin** = 0
- Column maxima: {1, 3} → **Minimax** = 1

Conclusion: Maximin (0) ≠ Minimax (1), so **no saddle point exists**. This game requires a mixed strategy.

5. Find the saddle point (if any) and the value of the game for the matrix:

$$\begin{pmatrix} -5 & 2 \\ -7 & -1 \end{pmatrix}$$

Solution:

	B ₁	B ₂	Row Min
A ₁	-5	2	-5
A ₂	-7	-1	-7
Col Max	-5 ↑	2	

- Row minima: {-5, -7} → **Maximin** = -5 (A plays A₁)
- Column maxima: {-5, 2} → **Minimax** = -5 (B plays B₁)

Conclusion: Maximin = Minimax = -5 ✓

Saddle Point: Exists at position (A₁, B₁)

Optimal Strategy:

- Player A plays A₁
- Player B plays B₁

Value of the Game: $V = -5$

Interpretation: This game favors Player B. If both play optimally, Player A loses ₦5 per game (equivalently, Player B wins ₦5 per game).

ASSIGNMENT (WEEK 8)

1. Find the optimal mixed strategies (p_1, p_2, q_1, q_2) and the value (V) for the game in Evaluation Q4, which has no saddle point.

Given matrix: $M = \begin{pmatrix} 1 & 0 \\ -2 & 3 \end{pmatrix}$

where $a = 1, b = 0, c = -2, d = 3$

Find Player A's Strategy:

Row differences:

- $d_1 = |a - b| = |1 - 0| = 1$
- $d_2 = |c - d| = |-2 - 3| = |-5| = 5$

Probabilities: $p_1 = \frac{d_2}{d_1 + d_2} = \frac{5}{1 + 5} = \frac{5}{6} \approx 0.833$

$$p_2 = \frac{d_1}{d_1 + d_2} = \frac{1}{1 + 5} = \frac{1}{6} \approx 0.167$$

Player A's Strategy: Play A_1 about 83.3% of the time, play A_2 about 16.7% of the time.

Find Player B's Strategy:

Column differences:

- $d^c_1 = |a - c| = |1 - (-2)| = |3| = 3$
- $d^c_2 = |b - d| = |0 - 3| = |-3| = 3$

Probabilities: $q_1 = \frac{d^c_2}{d^c_1 + d^c_2} = \frac{3}{3 + 3} = \frac{3}{6} = \frac{1}{2} = 0.5$

$$q_2 = \frac{d^c_1}{d^c_1 + d^c_2} = \frac{3}{3 + 3} = \frac{3}{6} = \frac{1}{2} = 0.5$$

Player B's Strategy: Play B_1 50% of the time, play B_2 50% of the time.

Find the Value of the Game:

$$V = \frac{ad - bc}{a + d - (b + c)}$$

$$V = \frac{(1)(3) - (0)(-2)}{1 + 3 - (0 + (-2))}$$

$$V = \frac{3 - 0}{4 - (-2)} = \frac{3}{6} = \frac{1}{2} = 0.5$$

Complete Answer:

- $p_1 = 5/6, p_2 = 1/6$ (Player A plays A_1 83.3% of the time, A_2 16.7% of the time)
 - $q_1 = 1/2, q_2 = 1/2$ (Player B plays B_1 50% of the time, B_2 50% of the time)
 - $V = 0.5$ (Player A wins ₦0.50 per game on average)
-

2. Players A and B play a game of "matching pennies." If the coins match (HH or TT), A wins ₦10 from B. If they don't match (HT or TH), B wins ₦10 from A.

(a) Construct the payoff matrix for Player A.

Solution:

Let's define strategies:

- H = Show Heads
- T = Show Tails

The payoff matrix (for Player A):

	B: Heads	B: Tails
A: Heads	10	-10
A: Tails	-10	10

Explanation:

- (H, H): Coins match → A wins ₦10
 - (H, T): Coins don't match → A loses ₦10 (B wins ₦10)
 - (T, H): Coins don't match → A loses ₦10
 - (T, T): Coins match → A wins ₦10
-

(b) Does this game have a saddle point?

Solution:

	B: H	B: T	Row Min
--	-------------	-------------	----------------

	B: H	B: T	Row Min
A: H	10	-10	-10 ←
A: T	-10	10	-10 ←
Col Max	10	10	

- Row minima: $\{-10, -10\} \rightarrow \mathbf{Maximin} = -10$
- Column maxima: $\{10, 10\} \rightarrow \mathbf{Minimax} = 10$

Conclusion: Maximin $(-10) \neq$ Minimax (10) , so **no saddle point exists**.

(c) What is the optimal strategy for both players and what is the value of the game?

Solution:

$$\text{Matrix: } M = \begin{pmatrix} 10 & -10 \\ -10 & 10 \end{pmatrix}$$

where $a = 10$, $b = -10$, $c = -10$, $d = 10$

For Player A:

Row differences:

- $d_1 = |10 - (-10)| = |20| = 20$
- $d_2 = |-10 - 10| = |-20| = 20$

$$p_1 = \frac{20}{20 + 20} = \frac{20}{40} = \frac{1}{2}$$

$$p_2 = \frac{20}{20 + 20} = \frac{20}{40} = \frac{1}{2}$$

For Player B:

Column differences:

- $d^c_1 = |10 - (-10)| = |20| = 20$
- $d^c_2 = |-10 - 10| = |-20| = 20$

$$q_1 = \frac{20}{20 + 20} = \frac{20}{40} = \frac{1}{2}$$

$$q_2 = \frac{20}{20 + 20} = \frac{20}{40} = \frac{1}{2}$$

Value of the Game:

$$V = \frac{(10)(10) - (-10)(-10)}{10 + 10 - (-10 + (-10))}$$

$$V = \frac{100 - 100}{20 - (-20)} = \frac{0}{40} = 0$$

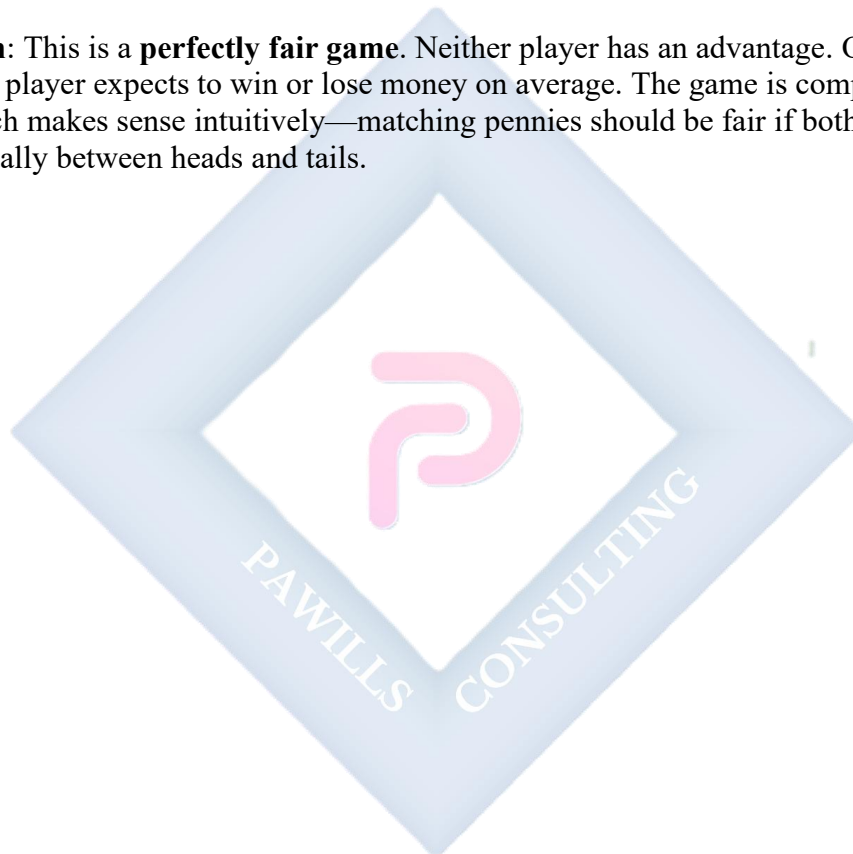
Complete Answer:

Optimal Strategy for Both Players:

- Each player should show Heads 50% of the time and Tails 50% of the time
- The strategy is perfectly symmetric

Value of the Game: $V = 0$

Interpretation: This is a **perfectly fair game**. Neither player has an advantage. Over many games, neither player expects to win or lose money on average. The game is completely balanced, which makes sense intuitively—matching pennies should be fair if both players randomize equally between heads and tails.



WEEK 9: REVISION

This week consolidates all major topics from Weeks 1-8. Below is a comprehensive review guide with key concepts, formulas, and strategies.

1. INTEGRATION

Key Formulas to Master

Basic Power Rule: $\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

The $(ax + b)^n$ Rule: $\int (ax + b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C \quad (n \neq -1)$

Logarithmic Rule (when $n = -1$): $\int \frac{1}{x} dx = \ln|x| + C$

General Log Rule: $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + C$

Definite Integrals

$$\int_a^b f(x) dx = F(b) - F(a)$$

where $F(x)$ is the antiderivative of $f(x)$.

Example: $\int_1^3 (2x + 1) dx = [x^2 + x]_1^3 = (9 + 3) - (1 + 1) = 12 - 2 = 10$

Applications of Integration

1. Kinematics (Motion)

The relationship: **Acceleration** $\rightarrow$ **Velocity** $\rightarrow$ **Displacement**

$$\int a \rightarrow v \rightarrow s$$

Example: If acceleration $a(t) = 6t \text{ m/s}^2$, and at $t = 0$, $v = 2 \text{ m/s}$:

$$v(t) = \int 6t dt = 3t^2 + C$$

Using initial condition: $v(0) = 2 \rightarrow C = 2$

$$v(t) = 3t^2 + 2 \text{ m/s}$$

2. Area Under a Curve

$$A = \int_a^b y \, dx$$

Example: Find the area under $y = x^2$ from $x = 0$ to $x = 2$:

$$A = \int_0^2 x^2 \, dx = \left[\frac{x^3}{3} \right]_0^2 = \frac{8}{3} - 0 = \frac{8}{3} \text{ square units}$$

3. Trapezoidal Rule (Numerical Integration)

When exact integration is difficult:

$$\int_a^b y \, dx \approx \frac{h}{2} [y_0 + 2(y_1 + y_2 + \dots + y_{n-1}) + y_n]$$

where $h = \frac{b-a}{n}$ (the width of each strip)

Example: Estimate $\int_0^2 \sqrt{1+x^2} \, dx$ using 4 strips.

$$h = \frac{2-0}{4} = 0.5$$

x-values: 0, 0.5, 1, 1.5, 2

x	0	0.5	1	1.5	2
y	1	1.118	1.414	1.803	2.236

$$A \approx \frac{0.5}{2} [1 + 2(1.118 + 1.414 + 1.803) + 2.236] \quad A \approx 0.25[1 + 2(4.335) + 2.236] = 0.25 \times 11.906 \approx 2.977$$

4. Volume of Revolution (Rotating about x-axis)

$$V = \pi \int_a^b y^2 \, dx$$

Example: Find the volume when $y = x$ is rotated about the x-axis from $x = 0$ to $x = 3$:

$$V = \pi \int_0^3 x^2 \, dx = \pi \left[\frac{x^3}{3} \right]_0^3 = \pi \cdot \frac{27}{3} = 9\pi \text{ cubic units}$$

2. CONIC SECTIONS

The Four Conics

1. **Circle:** Set of all points equidistant from a center
2. **Parabola:** Set of all points equidistant from a focus and directrix
3. **Ellipse:** Set of all points where sum of distances to two foci is constant
4. **Hyperbola:** Set of all points where difference of distances to two foci is constant

Parabola (Vertex at Origin)

Four Standard Forms:

Equation	Opens	Focus	Directrix
$y^2 = 4ax$	Right	$(a, 0)$	$x = -a$
$y^2 = -4ax$	Left	$(-a, 0)$	$x = a$
$x^2 = 4ay$	Up	$(0, a)$	$y = -a$
$x^2 = -4ay$	Down	$(0, -a)$	$y = a$

Example: For $y^2 = 12x$:

- $4a = 12 \rightarrow a = 3$
- Opens right
- Focus: $(3, 0)$
- Directrix: $x = -3$

Ellipse

Standard Form (center at origin):

Horizontal major axis: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where $a > b$

Vertical major axis: $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$ where $a > b$

Key Properties:

- **Vertices:** $(\pm a, 0)$ or $(0, \pm a)$
- **Co-vertices:** $(\pm b, 0)$ or $(0, \pm b)$
- **Foci:** $(\pm c, 0)$ or $(0, \pm c)$ where $c^2 = a^2 - b^2$
- **Eccentricity:** $e = \frac{c}{a}$ where $0 < e < 1$

Example: $\frac{x^2}{25} + \frac{y^2}{9} = 1$

- $a^2 = 25 \rightarrow a = 5$, $b^2 = 9 \rightarrow b = 3$

- Horizontal major axis
- Vertices: $(\pm 5, 0)$
- $c^2 = 25 - 9 = 16 \rightarrow c = 4$
- Foci: $(\pm 4, 0)$
- Eccentricity: $e = 4/5 = 0.8$

Hyperbola

Standard Form (center at origin):

Horizontal transverse axis: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Vertical transverse axis: $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

Key Properties:

- **Vertices:** $(\pm a, 0)$ or $(0, \pm a)$
- **Foci:** $(\pm c, 0)$ or $(0, \pm c)$ where $c^2 = a^2 + b^2$
- **Asymptotes** (horizontal): $y = \pm \frac{b}{a}x$
- **Eccentricity:** $e = \frac{c}{a}$ where $e > 1$

Example: $\frac{x^2}{16} - \frac{y^2}{9} = 1$

- $a^2 = 16 \rightarrow a = 4, b^2 = 9 \rightarrow b = 3$
- Horizontal transverse axis
- Vertices: $(\pm 4, 0)$
- $c^2 = 16 + 9 = 25 \rightarrow c = 5$
- Foci: $(\pm 5, 0)$
- Asymptotes: $y = \pm(3/4)x$
- Eccentricity: $e = 5/4 = 1.25$

3. STATICS

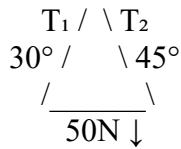
Resolution of Forces Method

Steps:

1. Draw a Free Body Diagram (FBD)
2. Resolve all forces into components (horizontal and vertical)
3. Apply equilibrium conditions:
 - $\Sigma F_x = 0$ (sum of horizontal forces = 0)
 - $\Sigma F_y = 0$ (sum of vertical forces = 0)

4. Solve the equations

Example: A 50 N weight hangs from two strings making angles of 30° and 45° with the horizontal. Find the tensions T_1 and T_2 .



Horizontal equilibrium: $T_1 \cos 30^\circ = T_2 \cos 45^\circ$

Vertical equilibrium: $T_1 \sin 30^\circ + T_2 \sin 45^\circ = 50$

Solving: $T_1 \approx 38.3 \text{ N}$, $T_2 \approx 46.9 \text{ N}$

Principle of Moments

For a body in equilibrium:

Sum of Clockwise Moments = Sum of Anti-Clockwise Moments

$$\sum M_{CW} = \sum M_{ACW}$$

Moment = Force $\times$ Perpendicular Distance

Example: A 4 m uniform beam of weight 100 N is supported at one end. A 200 N load is placed 3 m from the support. Find the support reaction.

Taking moments about the support: $R \times 0 = 100 \times 2 + 200 \times 3$ $0 = 200 + 600$

This is incorrect. We need vertical equilibrium: $R = 100 + 200 = 300\text{N}$

Friction

Laws of Friction:

1. Friction opposes motion
2. $f \leq \mu N$ (friction $\leq$ coefficient $\times$ normal reaction)
3. **Maximum friction:** $f = \mu N$ (limiting friction)

On a Horizontal Surface:

- Normal reaction: $N = W$ (weight)
- Friction force: $f = \mu N = \mu W$

On an Inclined Plane (angle θ):

- Normal reaction: $N = W \cos \theta$

- Friction force: $f = \mu N = \mu W \cos \theta$
- Component down slope: $W \sin \theta$

Example: A 20 kg block rests on a surface with $\mu = 0.3$. What horizontal force is needed to move it?

$$F = f_{\max} = \mu N = 0.3 \times (20 \times 10) = 0.3 \times 200 = 60\text{N}$$

4. MATHEMATICAL MODELLING

The Modelling Cycle (Quick Review)

1. **Identify** real-world problem
2. **Make assumptions** & define variables
3. **Construct** mathematical model
4. **Solve** the model
5. **Interpret** & validate results

Key Tip: Always state your assumptions clearly!

Independent vs. Dependent Variables

- **Independent (x):** What you control or changes naturally
- **Dependent (y):** What you measure or observe

Model form: $y = f(x)$

5. GAME THEORY

Quick Reference

Saddle Point Method (Pure Strategy):

1. Find row minima $\rightarrow$ take maximum (Maximin)
2. Find column maxima $\rightarrow$ take minimum (Minimax)
3. If Maximin = Minimax $\rightarrow$ Saddle point exists
4. Value of game = saddle point value

Mixed Strategy (2×2 , no saddle point):

For matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$p_1 = \frac{|c - d|}{|a - b| + |c - d|}$$

$$q_1 = \frac{|b - d|}{|a - c| + |b - d|}$$

$$V = \frac{ad - bc}{a + d - (b + c)}$$

REVISION TIPS

Integration

✓ Practice all types: power rule, $(ax+b)^n$, logarithmic ✓ Master definite integrals ✓ Review all applications: kinematics, area, trapezoidal rule, volume

Conic Sections

✓ Memorize standard forms for all four conics ✓ Remember: $c^2 = a^2 - b^2$ (ellipse), $c^2 = a^2 + b^2$ (hyperbola) ✓ Practice finding foci, vertices, eccentricity

Statics

✓ Always draw Free Body Diagrams ✓ Know when to use resolution vs. moments ✓ Don't forget: $f = \mu N$ for friction problems

Modelling

✓ State assumptions clearly ✓ Identify variables correctly ✓ Validate your model against reality

Game Theory

✓ Check for saddle points first ✓ Use mixed strategies only when no saddle point exists ✓ Remember: payoff matrix shows Player A's gains

PRACTICE PROBLEMS

1. Integration: Evaluate $\int_0^2 (3x^2 + 2x) dx$

2. Conic Section: Find the foci of $\frac{x^2}{36} + \frac{y^2}{20} = 1$

3. Statics: A 5 kg mass hangs from two strings at 60° to each other. Find the tensions.

4. Modelling: Model the cost C of running a taxi if fuel costs ₦200/km and driver wages are ₦5000/day.

5. Game Theory: Find the value of the game: $\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}$